

Plantar Pressure-Based Insole Gait Monitoring Techniques for Diseases Monitoring and Analysis: A Review

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Among diverse wearable techniques, insole-based plantar pressure monitoring systems have surged as a leading technology to monitor patient's chronic disease progression. Such technological feat has been made possible due to the strong correlation between gait and disease status. Hence, insole-based plantar pressure monitoring techniques are growing rapidly worldwide; with several research institutions and enterprises showing an increased interest in the field. This review intends to first explain the working principles of mainstream insole plantar sensing techniques and design considerations such as sensing material selection and electronics design requirements, and then the state-of-the-art algorithms for plantar pressure distribution reconstruction. Following, this article will discuss disease monitoring applications and the extraction of disease features. Finally, insight regarding common challenges and their potential solutions within the field would be elucidated.

energy expenditure^[9] can also be obtained for a more detailed assessment from insole system consisting of pressure sensors, inertial measurement unit, and EMG sensors. Both direct and indirect means require a large and highly accurate dataset of raw plantar pressure, which is used to be an impossible task, but now becomes possible with the advanced services provided by recently fast developed wearable systems,^[10–12] including no restriction on the testing environment, minimal nuance with the user's daily activities during data collection, and sufficient data from long-term monitoring for disease analysis and diagnosis. Hence, insole plantar pressure monitoring techniques are garnering global attention.

Fruitful research outcomes on insole plantar pressure sensing prototype devel-

1. Introduction

Gait, the comprehensive result of the coordination of the central nervous system, peripheral nerves, and multiple skeletal muscle groups of the lower limbs, is considered to be one of the most critical patterns in human daily activities.^[1] By analyzing gait patterns, parameters with physiological explanations can be obtained for clinical diagnosis and rehabilitation. Among all parameters (e.g., plantar pressure, joint angle, and electromyography (EMG) signal), plantar pressure presents various diseases such as Parkinson's disease (PD), diabetes, and stroke.^[2–8] Besides, with direct means (e.g., summarization, weighted average, threshold division) and indirect means (e.g., regression model and artificial neural network), other features like

opments and associated data analysis have been presented by medical professionals and engineers since the last decade, as shown in **Figure 1**. Nevertheless, the success of a broad use of insole systems for patients has not been reported yet. By surveying researchers in the field and reviewing the literature, a key factor to explain this phenomenon might be the knowledge gap^[13] between medical professionals and engineers. Under this circumstance, this paper is presented to bridge the knowledge gap between them through five main aspects. First, working principles, material selection, and architectures of state-of-the-art insole force sensing techniques are reviewed in detail to provide readers with a guide when developing an insole system. Second, the design considerations of analog front end, based on different sensing mechanisms, is discussed for maximizing plantar pressure detection accuracy. Third, algorithms on high-resolution plantar stress reconstruction and gait parameter extraction are explained. Fourth, the applications and demonstration of the algorithms utilized in disease analysis and symptom evaluation are presented. Fifth, challenges and potential developing trends of current insole techniques are discussed, which may inspire colleagues to promote the development of the field.

This article is structured as follows: Section 2 presents the sensing mechanisms, material selection and sensor constructions of plantar pressure detection for instrumented shoes. The electronic design consideration for developing insole plantar pressure monitoring systems is discussed in Section 3. Section 4 reviews the algorithms used to compute

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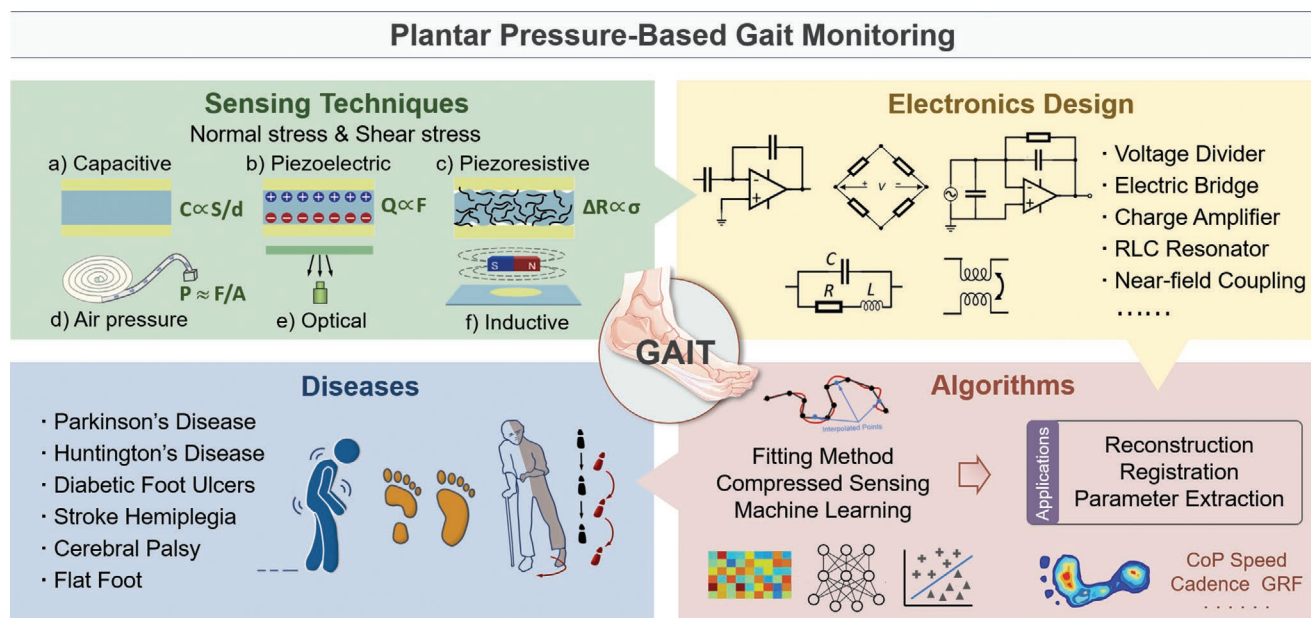


Figure 1. An overview of insole plantar pressure monitoring techniques and applications in diseases analysis.

gait parameters, including reconstruction and registration of the plantar pressure distribution. Section 5 reviews the gait detection system with multisensory fusion and its methods for extracting gait parameters. In Section 6, the algorithms for identifying symptoms or diseases based on gait parameters are reviewed. Section 7 then discusses the current challenges and future trends in this field. Finally, conclusions are drawn in Section 8.

2. Insole Pressure Sensor Constructions

Insole-based pressure sensors are able to obtain a variety of valuable information on human gaits, such as human balance, lower limb articular movement, and walking phase. Utilizing these temporal and spatial characteristics acquired by insole sensors, diagnosis and rehabilitation evaluation for some motor and neurological diseases can be performed. To provide high-quality data for disease analysis, insole sensors with high detection accuracy are required, giving rise to some general sensor design rules, after extensive experiments were carried out:

- 1) No or little impact is given to the user's natural gaits.^[14] Generally, small volume, lightweight, and good flexibility are desired attributes for insole devices.
- 2) The force sensing range, device responsivity's long-term stability under extreme conditions (such as heavy loads, frequent bends, and changing temperature), indicating the material preparation device design are of significance.
- 3) The layout of sensors should match the features of diseases under interest.
- 4) The system's scanning speed should support specific application requirements.

With the above design rules, numerous insole architectures for disease detection have been proposed in the literature. Some

are even commercialized and put into practice use. In this section, these architectures' working principle, sensing materials and device architectures will be detailed explained, together with sensor deployment considerations.

2.1. Sensing Mechanisms, Material Selection, and Device Architectures

Traditionally utilized insole plantar force sensing mechanisms include piezoresistive, resistive, capacitive, piezoelectric, and inductive means. In recent years, optical and air pressure-based techniques emerge and bring promising results, triggering research interests. During the design process of an insole sensor system, the most significant consideration factor is the application scenario, for example, stroke rehabilitation or diabetes early-stage detection. For most application scenarios, if not all, key parameters to consider in developing prototypes are pressure sensing range, hysteresis, sensor volume, and flexibility.^[15] Besides, the materials of sensing devices will also significantly affect the patient's wearing experience and the sensing performance, which will determine the accuracy of the detected plantar pressure information and affect the effectiveness of subsequent algorithms.

With the analysis of the human walking process and the results of existing research work, we summarize the basic requirements for materials and device architectures used in plantar pressure detection as follows^[15–19]: 1) mechanical properties: flexible material rather than brittle material is suitable for frequent bending during walking; 2) full sensing scale: the upper limit of normal stress range is widely agreed to be over 1000 kPa in walking analysis, 1900 kPa in daily activities, and 3000 kPa in some extreme situations^[15,16]; 3) pressure resolution: <10 kPa is a widely accepted index for insole applications (e.g., abnormal pattern detection for PD) based on plantar

pressure sensors^[17]; For detecting the pressure abnormality in associated regions of some specific diseases (e.g., diabetes, rheumatoid arthritis (RA), flat foot and cavus foot), the pressure resolution should be <1 kPa; 4) pressure detection error: less than 1% full scale; 5) measurement bandwidth: 100 Hz for walking, and 200 Hz for vigorous exercises (e.g., running and jumping); 6) compatibility for both static measurement and dynamic measurement; 7) hysteresis: less than 5% full scale; 8) stability: error $< 5\%$ full scale for each sensor after more than ten thousands loading-unloading cycles; 9) shear pressure detection is expected for some specific diseases, such as patients with diabetes.^[19]

In what follows, we overview six major insole pressure sensing mechanisms, and corresponding material selection and device architectures in disease detection applications.

2.1.1. Piezoresistive Techniques

The resistive-based technique, which establishes a relationship between applied pressure and changed electric resistivity of the device, is widely utilized in insole pressure sensing due to its simple structure and high force sensitivity. Early studies paste individual commercial force sensitive resistor (FSR) sensors on specific insoles' positions to conduct the measurement.^[20–24] For example, as Figure 2a shows, 12 FSRs are placed in the areas under toes, forefoot, midfoot, and heels to analyze joint torque anomaly in stroke hemiplegia patients^[20]; the reasons for this location setting will be provided in the following subsection. A flexible circuit board is designed as the substrate to solder the leads, and the sensors are then attached by adhesive. S. Bamberg et al.^[23] presented an insole pressure system for balance ability assessment of patients with PD, where 16 FSRs are affixed to the forefoot and heel areas. The insole is enclosed

in shrink-wrap to protect the sensors from motion-induced abrasion and displacement.

Commercial utilized piezoresistive materials suitable for insole plantar detection contains conductive ink,^[25,26] conductive fabric^[27] and liquid metals.^[28–31] Among them, conductive ink (filler contains metal nanoparticles, carbon particles, polymer, etc.) is the most widely used piezoresistive category in plantar pressure detection due to its simple structure and low cost. Conductive fabric (made from gold, carbon, titanium, nickel, silver, or copper on the base fabric materials like cotton, wool, polyester, and nylon) can be easily integrated with wearable clothing like shoes and shirts, ensuring the users' comfort while sensing. Besides, pressure sensors based on liquid metal (e.g., EGaIn^[31]) provide high stability for long-term experiments, simple structure and fabrication process, and can easily be integrated in the wearable systems. However, commercial utilized piezoresistive materials have some challenges to be addressed for insole applications: first, the conduct between the conductive ink layer will be broken under repeated bending and result in the problem of long stability^[32]; second, the connection between the conductive fabric and the analog front end is not firm, and the connection is easily affected by mechanical chafing during applications, resulting in the contact resistance increase and even fracture at the connection interface^[30]; third, the liquid metal track needs to be packaged and protected in the elastomer substrate. Under the combined effect of its high viscosity and the elasticity of the elastomer substrate, the pressure sensor based on the liquid metal will produce a considerable hysteresis effect (approximately 50% delay of signal at lower force speed ≈ 2 mm min⁻¹),^[31] which is not suitable for detecting plantar pressure signals with rapid change.

To enhance piezoresistive sensing performance, the method of utilizing nanocomposite^[33–35] or adding nanostructure^[36–38] has aroused researchers' interest. Multiwalled carbon nanotube

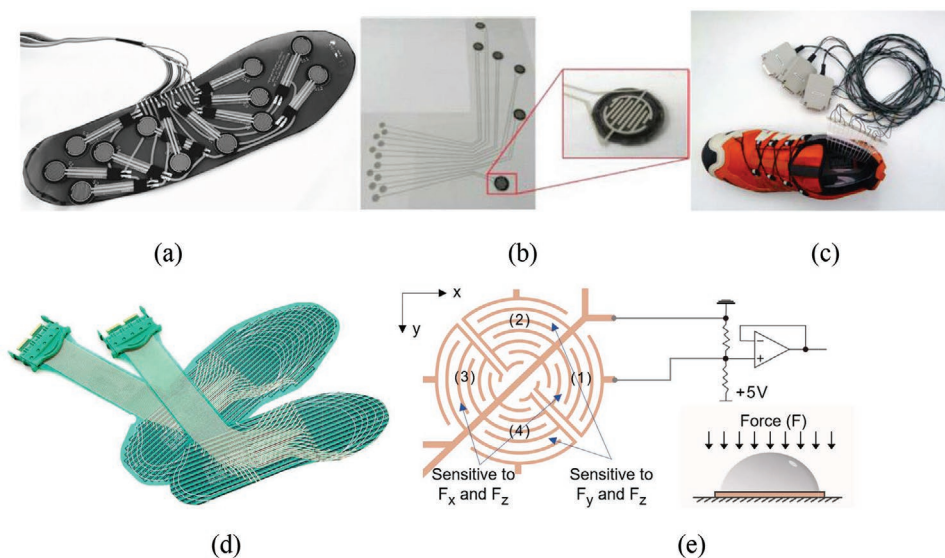


Figure 2. a) The insole presented in ref. [24], including a flexible circuit board and 12 FSR sensors. Reproduced with permission.^[24] Copyright 2009, IEEE. b) The printed insole and (c) system implement. Reproduced with permission.^[49] Copyright 2016, IEEE. d) F-Scan insole sensor (Tekscan, USA). Reproduced with permission.^[42] Copyright 2021, Tekscan Inc. e) Conceptual view of the piezoresistive-based 3D force sensor. Reproduced with permission.^[48] Copyright 2008, IEEE.

(MWCNT)—Polydimethylsiloxane (PDMS)^[34,35] is a commonly used nanocomposite in the region of human–machine interactivity for its strength and flexibility, and can provide and a wider detection range (2.5–640 kPa^[35]).

By introducing microstructures in the sensing material layer^[36] or on the substrate,^[37,38] the deformation of the sensor with loading can be increased, hence reaching an extremely high sensitivity ($\approx 15.1 \text{ kPa}^{-1}$ with micro dome film array composed of carbon nanotubes and polydimethylsiloxane prepolymer^[27]). Besides, microstructure provides a better mechanical property, resulting in a faster response time (40 ms for loading and 60 ms for unloading with two vertically aligned carbon nanotubes on PDMS substrates^[37]).

During the past decades, the technique of integrating the piezoresistive material (e.g., carbon nanotube^[39]) into transistors to fabricate electrical pressure-sensing components with an electrical output of source-drain stage current (I_D) has been widely used. This technique can reduce the sensor volume with low environmental noise interference, improve device integration, and simplify the electrical design, hence it is becoming a new research hotspot.^[39,40]

The previously mentioned adhesive-based methods often lead to the overlap of sensor elements and lead-out lines, especially when a large number of sensors are required, causing unstable mechanical capacity and user's discomfort. To address this issue, screen printing technology can be used to perform whole-piece design and fabrication. For instance, such a process is used to prepare piezoresistive material and electrode array layers in ref. [41]. The printed insole and the system implementations are shown in Figure 2b,c; its capability in plantar pressure sensing is verified in the detection of plantar pressure ulcers. The piezoresistive technique has been utilized in many commercial insole products. The F-Scan (Tekscan, USA) system^[42,43] (shown in Figure 2d), with a sensing range of 7–1034 kPa, is extensively used for various clinical applications, such as the analysis of RA^[44] and pain pertaining to foot pathology.^[45] The SurroSense Rx Insole, developed by Orpyx Medical Technologies Inc., is designed explicitly for early warning of diabetes. When the detected plantar pressure at a particular location is above a predetermined threshold ($>450 \text{ N cm}^{-2}$), the user will be alarmed about the potential risk of suffering a foot ulcer. In the study of ref. [46], F. Melakessou et al. developed a resistive insole device for peak pressure detection and activity tracking for diabetic patients. In each sensor cell, the upper resistive layer will deflect toward the lower layer under stress; once the contact is established, the contact area increases due to a rise in external pressure. The structure has a pressure range of 25–700 kPa, with an unrepeatability of 15% over the lifetime.

Besides, strain gauges attached on mechanical structures can measure the strain under stress. For instance, R. Lind et al.^[47] developed a mechanical insole for multi-axis force-sensing utilizing strain gauge load cells, which achieves a high vertical load capacity of 5.34 kN and a shear load capacity of 890 N; however, the rigid material cannot be bend as strong as standard soles, and the heavy weight (0.8 kg) brings inconvenience to users.

In addition to normal stress sensing, the resistive techniques with smartly designed structures can detect shear stress. For instance, as Figure 2e shows, an elastic hemisphere (diameter:

10 mm) made of pressure sensitive electric conductive rubber is fabricated on four interdigital electrodes in a fan shape.^[48] The resistance changes caused by normal stress have the same response at each of the four electrodes, while the shear force causes opposite changes at the diagonal electrodes. A. Amemiya et al.^[49] designed an insole system with four piezoresistive cantilever beams-based commercial MEMS triaxial force sensors (dimension: 10 mm $\times$ 10 mm $\times$ 1.3 mm) to detect stress at the upper detection limit of 400 kPa for normal stress and 80 kPa for shear stress, respectively. However, the structures for shear force sensing are relatively complex and have not been commercialized yet.

2.1.2. Capacitive Techniques

The capacitive sensors detect the applied force by estimating the distance change between the upper and lower electrodes of an elastomer medium. This technique has the advantage of high force sensitivity and good dynamic performance; However, its force range is limited, and the sensor is sensitive to humidity and electromagnetic interference (EMI).

The basic structure of a capacitive pressure sensor can be divided into two layers: the dielectric layer and the lead-out electrode layer. Parallel plate capacitor is the most widely used structure.^[50–52] The conventional dielectric materials contain air, plastic, glass, silicon, ceramic, polydimethylsiloxane,^[52] etc., and the conventional materials utilized in lead-out electrode layer contains copper/silver tracks, indium tin oxide (ITO) conductive glass and polyethylene terephthalate (PET)/PDMS substrates. However, the pressure-sensing capacitor with parallel plate structure suffers from small capacitance change and poor sensor sensitivity due to the material mechanical modulus.^[51]

Two methods are widely utilized to boost the sensitivity of capacitive sensors. One is that dielectric materials of high dielectric constants can be used, like MWCNT and Sylgard 184 PDMS.^[53]

Another method is to integrate microstructures into capacitive sensor layers, which can improve the dielectric layer's effective permittivity and reduce the distance of lead-out electrode layers when pressing the sensor.^[54–56] In ref. [54], S. Masihi et al. utilize an additive screen-printing processes to fabricate the porous PDMS by mixing PDMS with NaHCO_3 and HNO_3 to produce a dielectric layer, which is capable of achieving a high sensitivity of about 0.5 kPa^{-1} in the sensing range of 0–900 kPa. In ref. [56], a 15 μm thickness PDMS dielectric layer is built using a pyramid microstructure with a detection range of 0–600 kPa and a sensitivity of 0.021 kPa^{-1} . Besides, the pyramid microstructure adjusts the contact mode between the dielectric layer and the electrode layer, reducing the viscoelastic deformation-related problems, which allows no response delay in insole applications.

In recent years, integrating the capacitive sensor into the field-effect transistor and setting the sensor as the gate capacitance to adjust the drain current has exposed the advantages of system integration improvement, complexity reduction of the electronic system, a fast response speed ($\approx 50 \text{ ms}$), an extremely high sensitivity ($\approx 8.4 \text{ kPa}^{-1}$), a wide detection range (0–2 MPa) and an immunity to electromagnetic interference.^[57–59] The

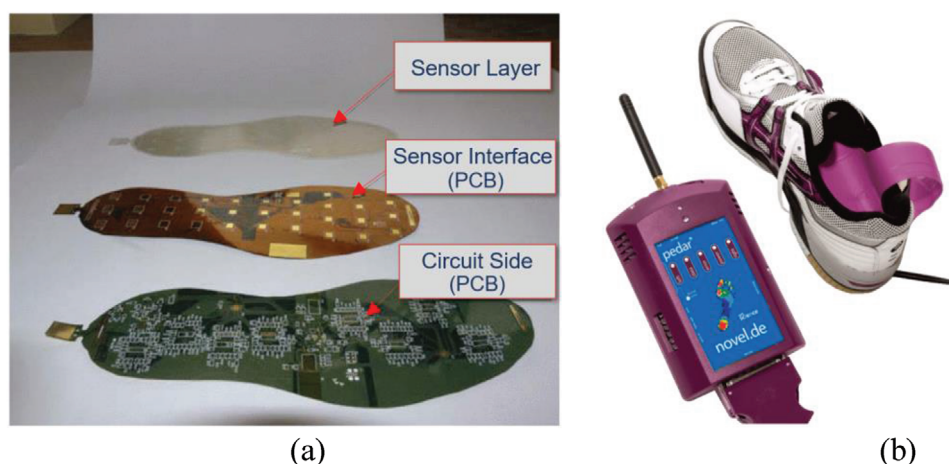


Figure 3. a) The capacitive-based normal stress sensor and flexible circuit board. Reproduced with permission.^[60] Copyright 2015, IEEE. b) The commercial capacitive insole system Pedar-x. Reproduced with permission.^[61] Copyright 2021, Pedar.

typical configuration of integrated pressure-sensing gate capacitance contains polydimethylsiloxane microstructure on the ITO conductive glass and PET substrates^[57,58] and an air dielectric with amorphous oxide channels.^[59] However, the high cost and strict fabrication process condition prevent its promotion to wide-scale implementation and manufacturing.

In practical applications, Y. Mustafa et al.^[60] presented an insole pressure sensor based on a capacitance array with a sensing range of 15–1000 kPa for tracking stroke patients' rehabilitation progress. As shown in **Figure 3a**, the insole system consists of sensors layers and a flexible circuit layer, enabling the whole system integrated in-shoe.

Commercial capacitive insole products include the OpenGo (Moticon GmbH, Munich) and Pedar-x insole system^[61] (Nove Electronics, Germany) as shown in **Figure 3b**. The former is utilized for examining the weight-bearing gait pattern after extremity fractures.^[62] The system can work within the pressure range of 2.5–400 kPa with a lifetime of 100-km running. The Pedar-x system is widely-used in diabetic foot ulcers^[63] and ankle osteoarthritis diagnosis.^[64] Its sensing range can be chosen between 15–600 kPa and 30–1200 kPa, and the unrepeatability is tested to be 10% for an average of 11.8 days in daily life.^[65]

2.1.3. Piezoelectric Techniques

Piezoelectricity is the electric charge that accumulates on the surface of non-centrosymmetric materials in response to applied mechanical force. Insole piezoelectric sensors are widely assembled as electrode/piezoelectric film/electrode sandwiched structures, offering merits such as low power consumption, simple structure, and high sensitivity; therefore, they receive boosting research interests in recent years. However, the polarization orientation of piezoelectric materials cannot be perfectly aligned in the same direction, resulting in the sensitivity to unwanted force components. They are also unable to detect static force because leakage current occurs through the subsequent amplifying circuit. Hence, this technique has not been applied in commercial products yet.

Piezoelectric pressure sensors can be divided into three categories based on different materials: inorganic materials, organic

materials, and composite materials. Among inorganic materials, lead zirconate titanate ($\text{Pb}[\text{Zr}_{(x)}\text{Ti}_{(1-x)}]\text{O}_3$, PZT) is a widely used inorganic piezoelectric material in the detection of weak physiological pressure signals (e.g., blood pressure detection^[66]) because of its high sensitivity ($>50 \text{ pC N}^{-1}$). However, due to the brittle characteristics, PZT cannot be directly used in applications with frequent bending like insole plantar pressure detection. The mechanical properties can be improved by reducing its scale and mixing it with adhesive. In ref. [67], a pair of flexible screen-printed piezoelectric shoes were fabricated by printing a mixer of two PZT powders (2 and $0.8 \mu\text{m}$ PZT powder with mixing ratio of 82%:18%) and a polymer binder on alumina substrates and achieved a detection range larger than 700 N.

The organic piezoelectric materials show outstanding mechanical flexibility and mechanical durability.^[68–71] Its representative—polyvinylidene difluoride (PVDF) film is the most widely used piezoelectric material in plantar pressure detection. However, the low piezoelectric coefficient ($<30 \text{ pC N}^{-1}$) of PVDF hinders the further improvement of the sensor performance. Adding inorganic nanoparticles (e.g., surface-functionalized barium titanate (FD-BTO)/PVDF composite) into PVDF to form piezoelectric composites can significantly improve the piezoelectric properties, achieving a high sensitivity of 69.1 pC N^{-1} and the detection range of 0–500 kPa.^[72]

Besides, the combination of piezoelectric materials and transistors can reduce the coupled noise on the transmission path and improve the performance of the piezoelectric pressure sensor. It can also improve the integration of the electrical read-out system by directly outputting electrical quantities.^[73,74] Typical structures contain piezoelectric metal-insulator-metal (containing brass substrate, PZT piezoelectric layer, Ag top layer electrode) and a PZT square microarray. The sensitivity reaches $4.55 \times 10^{-3} \text{ kPa}^{-1}$ with the upper detection limit of 94.18 kPa.^[73]

Moreover, piezoelectric film-based technology is a feasible solution for insole plantar shear force or multiaxial forces detection. Since the piezoelectric effect can occur on the axes of both vertical and parallel to the film surface, of which the force-to-voltage coefficient depends on the polarization direction and strength, piezoelectric films with different coefficient matrices can be selected for measuring the force in distinct directions. Below we introduce some representative works.

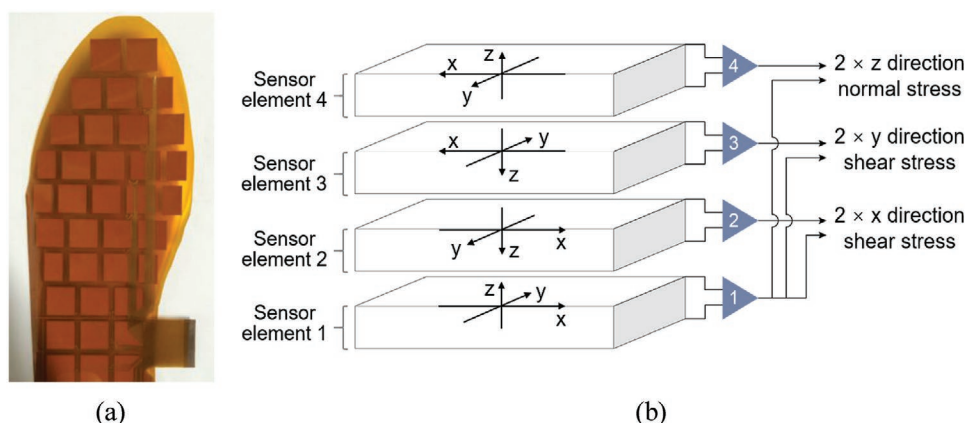


Figure 4. a) The piezoelectric-based smart insole with flexible printed circuit boards as electrode layers. Reproduced with permission.^[68] Copyright 2016, IEEE. b) Triaxial force sensor based on piezoelectric films.^[76]

E. Klimiec et al.^[71] developed a prototype with eight piezoelectric sensors for flatfoot and diabetic foot diagnosis. In order to minimize the crosstalk from other axials, rigid substrates are fixed on each sensor, experimentally allowing tacking of the eight points without distortion; however, the set of rigid structure will cause discomfort during walking. **Figure 4a** shows the smart insole proposed by F. Lin et al.,^[68] in which flexible printed circuit boards serve as the electrode layers. Such a structure is suitable for massive production, especially for a large sensor array.

The first piezoelectric based multiaxial force sensing insole was presented in 1995.^[75] In this design, three uniaxial sensors with different polarization directions are used to measure the normal and shear forces separately; however, the crosstalk from others is severe, which strongly limits the force sensing accuracy. To suppress force interferences, S. Kärki et al.^[76] stacked multiple films in opposite polarization orientations for neutralizing undesired charges. As shown in **Figure 4b**, each transducer consists of four independent commercial piezoelectric film sensors, both normal and shear forces are obtained by adding the output signals in pairs.

2.1.4. Inductive Techniques

Insole inductive force sensors include three components: a top conductive sheet, a supporting elastomer, and bottom coils for signal picking, where the coil is utilized to create an electromagnetic field surrounding the conductor. When

an external force relatively moves the three components, the source induction field will be changed, leading to sensor magnetoresistance variation.

The widely used materials for the top layer are metals and alloys with high electrical conductivity, for example, aluminum foil^[77] and mild steel.^[78] In some recent studies, conductive liquid is utilized to improve the sensor flexibility.^[79,80] For example, in ref. [79], the eGaIn liquid metal is injected into a female silicon rubber mold to form a soft metal layer. For the supporting elastomer, the ideal characteristics for insole applications include low creep, nonsignificant nonlinear effect, low stress relaxation, and high repeatability in cycles of high-pressure loading and unloading.^[81] Based on these considerations, the elastomer Smooth-Sil 950, EcoFlex 00–30, and Sylgard 184 are frequently chosen.^[80–82] The bottom coils are single- or multi-layer planar metal wires (e.g., copper wires, silver ink wires, and liquid eutectic alloys wires of Ga and In) on flexible substrates,^[83] fabricated by etching^[79,80,82] or printing technologies.^[84]

Based on inductive based pressure sensing principle, B. Nie et al.^[85] developed an insole pressure sensor, consisting of a top antenna layer, a fabric spacer, and a bottom ferrite film, as shown in **Figure 5a**. An interdigital capacitor is in series with the antenna to form an RLC network, whose resonant frequency is used to calculate the sensor magnetoresistance. The proposed sensor is flexible, with a dimensional size of 10 mm × 10 mm × 2.65 mm. However, it fails to measure pressure amplitudes above 20 kPa, which cannot meet some plantar pressure sensing scenarios requirements.

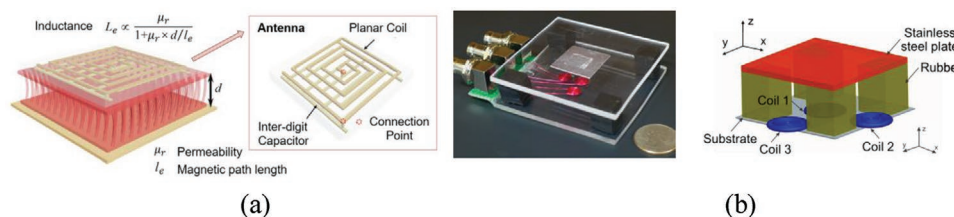


Figure 5. a) Schematic illustration of the inductive normal stress sensor, consisting of a top antenna, a bottom ferrite film, and an in-between fabric spacer. Reproduced with permission.^[85] Copyright 2019, Wiley-VCH. b) 3D electromagnetic induction sensor. Reproduced with permission.^[86] Copyright 2014, IEEE.

By measuring the horizontal displacement of the upper and lower plates, the shear stress can be estimated. For example, L. Du et al.^[86] presented an electromagnetic induction sensor for triaxial force sensing. As illustrated in Figure 5b, two parallel plates (a copper pad is attached to the top one, and three coils are glued to the bottom) are separated by four rubber blocks at the four corners. When an external force alters the two plates' spatial positions, the variation will be measured by the three spiral-wound coils. The system is capable of sensing normal and shear stress below 110.4 and 20.7 kPa, respectively. In this design, the large sensor size (76.2 mm × 76.2 mm × 22 mm) limits its use in high spatial-resolution measurement applications. The devices' large volume and rigidity may result in difficulties in shoe integration and affect natural gait.

2.1.5. Optical Techniques

Optical-based technology overcomes the problems of electromagnetic interference, hysteresis, and temperature drift, which are shared issues in the methods explained above. The optical sensors can be divided into fiber- and no fiber-based methods. The former is realized by observing light intensity attenuation or wavelength change induced by the deformation of fiber. These methods are with good integration capabilities; however, they suffer poor durability in plantar pressure detection, due to the fragile of optical fiber. The nonfiber based methods, based on the optoelectronic transduction principle, utilize the change of light intensity during light refraction and reflection to detect the deformation of sensor structure, and further calculate the external stress. The typical architecture is a flexible light-tight space with a fixed light source, an opaque light blocker, and a photodiode served as a light receiver. When an external force deforms the device, the blocker will displace and hinder a part of the light transmission; the photodiode will then detect the change of light intensity inside the space. For example, a sensor was integrated in this way for investigating gait abnormalities in amputees,^[87] where a middle curtain is utilized to close the light path between the light source and receiver. The total thickness of the insole is 5.7 mm, and the sensing range is 10–500 kPa. Due to the complex device structure and large sensor size, its applications are limited.

The structure of an optical-based force sensor is usually complex compared to the previous techniques. Optical sensors are constructed with three parts: luminous emission end,

transmitting media, and receiving end. The emitter is mainly divided in two kinds: light-emitting diode and laser diode. The light-emitting diode converts the forward voltage on a PN junction into light energy.^[88] The light color varies with the semiconductor material and doping composition. Laser diode is an optical maser that uses semiconductor materials as working substance, which mainly includes GaAs, CdS, InP, and ZnS.

The transmission module can be divided into two categories according to the medium: nonoptical fiber-based and optical fiber-based. For the former, some nontransparent elastic materials (including silicone, rubber, etc.) are added to the optical path to measure the optical path change caused by force-induced material deformation.^[87,89] In the latter category, silica glass was initially used in plantar force sensors owing to its wide availability.^[90] However, it suffers from serious fragility issues under human body load. In recent years, to solve this problem, multiple flexible materials have been used in the development of optical fibers, particularly including soft glass (e.g., As_2S_3 and As_2Se_3) and polymers (e.g., polymethyl methacrylate (PMMA), PDMS, and polyacrylamides (PAM) hydrogels).^[91,92] These materials allow higher robustness and wider force limit in insole applications.

Receiving end mainly includes photosensitive resistors and photosensitive transistors. The photosensitive resistor is a kind of semiconductor resistor, whose resistance is high in dark environments, but will significantly decrease under the light of a certain wavelength range. The most widely-used materials for photosensitive resistors are CdS, CdSe, PbSe, PbS, and CdTe. For photoelectric transistors, their input-output characteristics vary with environment illumination. The photosensitive materials used in photoelectric transistors include elemental semiconductors (Si, Ge) and compound semiconductors (GaAs, InAs, InSb, InGaAs). Photoelectric transistors are more preferred in plantar pressure measurement because of their fast response speed, low-temperature influence, and high sensitivity.

The optical sensing techniques also provide ways for multi-axial forces sensing. For instance, the insole triaxial force sensor proposed in ref. [89] uses reflected light intensity to recognize the proximity and tangential displacement of a reflective material, as illustrated in Figure 6a. Figure 6b shows the configuration of reflective optical sensors, of which the top black and white areas refer to absorptive and reflective materials, respectively. The total insole thickness is about 5.5 mm. Experimental results show mean errors of 68.1 N for normal force

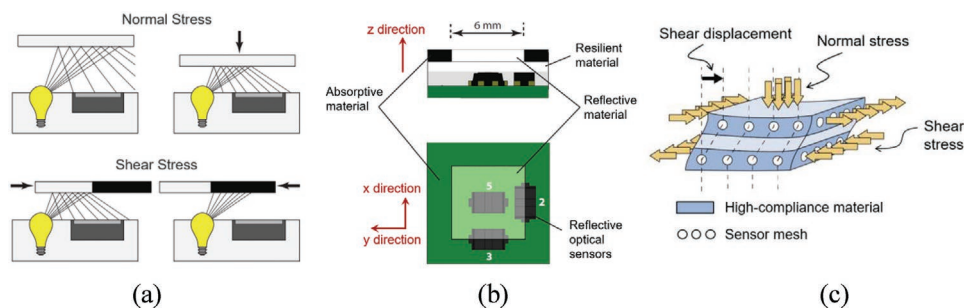


Figure 6. a) Principle of the reflective optical-based triaxial force sensors.^[89] b) Configuration of the triaxial force sensor.^[89] c) Fiber-optic based shear force sensor array.^[93]

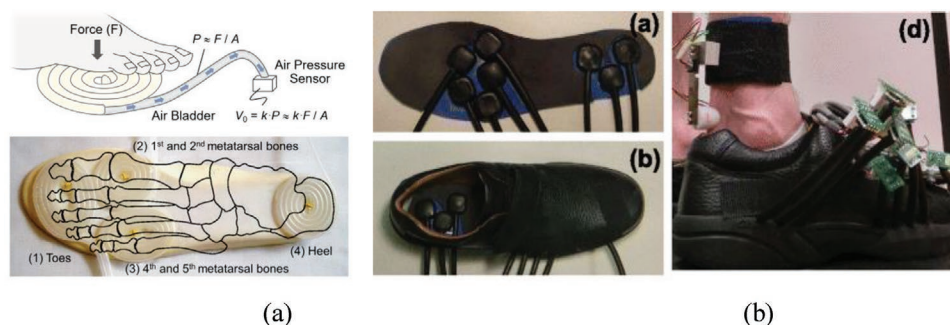


Figure 7. a) The working principle and device architecture of the air pressure sensors. Reproduced with permission.^[96] Copyright 2011, The American Society of Mechanical Engineers. b) An insole plantar pressure monitoring system based on eight custom neoprene air bladders. Reproduced with permission.^[101] Copyright 2015, BioMed Central.

sensing and 10.7 N for shear force sensing, around 10% of the corresponding force sensing range. In ref. [93], an optical fiber-based method for triaxial force detecting is proposed based on a sandwiched structure of two optical fiber arrays lying in rows and columns and an in-between elastomeric layer. As shown in Figure 6c, the normal stress distribution is obtained by measuring the light intensity loss of each fiber, and the shear force is interpreted by calculating the amount of pressure points misalignment between the two fiber mesh layers. The architecture is able to withstand a force of 1500 N in compression and 130 N in shear. In this design, the vertical spacing of crossed fibers at the same layer is around 1 cm, indicating an unpleasant thickness for wear; moreover, the structure encloses an intensive amount of fibers within a limited space, resulting in high device and circuitry complexity, high power consumption and low robustness. For simplifying the optic fiber-based sensor structure, C. Tavares et al.^[94] proposed a biaxial force sensor (9.0 mm × 16.0 mm × 5.5 mm) based on two fiber Bragg gratings with different central wavelengths and bracing structures. The applied normal and shear forces will stretch and compress the fibers, and the resultant deformation will lead to a wavelength shift. With the measured wavelengths and the prior knowledge of the two fibers' force-to-wavelength responsivity, the forces in both axes can be interpreted. In experiments, the estimated shear force can reach up to 350 N.

2.1.6. Air Pressure Based Techniques

An air pressure sensor is another feasible option which involves placing air pressure sensors under the user's feet. For air pressure-based force sensors, each unit includes a winded tube and an air pressure transducer; material of the former is crucial to the sensor performance. Widely-used materials for the tube are silicone, nylon, teflon, and polyurethane (PU).^[95] Among them, PU and teflon are with high hardness, thus not suitable for shoe interior environment. Nylon tubes have higher flexibility, hence they suffer from strong viscoelastic characteristics. In experiments, its creep effect under loading is 30% higher than that of silicone tubes.^[95] Therefore, silicone is the frequent-chosen material for insole air pressure.^[95,96]

The most widely implemented method is utilizing air bladders. B. Abinaya et al. set three air-pressure based sensing units (each contains air bladder and air pressure sensor, weights 20

g) to detect the abnormal motion phase with the fuzzy logic.^[97] Figure 7a shows the baroreceptor insole sensor developed by J. Bae et al.,^[96] containing four coiled air bladders (area: 4 × 4 cm²) and commercial air pressure transducers. When the bladders are deformed, the inside air pressure increases proportionally. The proposed sensor achieves an upper sensing range at 860 kPa. In dynamic force loading experiments, large hysteresis (approximately 16% at 1 Hz) caused by the air bladder's viscoelastic characteristics is observed. To compensate for this effect, the authors added a controlling unit to regulate the time delay, reducing the hysteresis to less than 1%.^[98] After manufacturing process improvement, H. Choi et al. stated that the room-temperature-vulcanizing silicone rubber KE-12 (from Shin-Etsu Chemical) can achieve a high sensing performance of 2.5 MPa dynamic range and 3.8% linearity. H. Choi et al. also presented an insole-type ground reaction force sensor system with four insole air-pressure-based sensing units. The ground reaction force is validated with the average root mean square errors (RMSEs) of 8.77% body weight by comparing it with the gold standard method (FP).^[99] However, the force from shear directions (anterior-posterior direction and medial-lateral direction) also leads to fluid volume change, which results in measurement errors of vertical plantar pressure.

One solution is to provide a vertical support sensing structure, thus removing the volume change caused by the shear stress. K. Kanghyun et al. presented a soft force-sensing pad (area: ø 5 cm), in which the air-filled structure is vertically supported by an elastomeric pillar array with a sealing cover. The vertical force will reduce the sensor units' volume, but the sealing cover holds back the deformation of shear directions, releasing the effect of shear force. Hence, the measurement result is merely relevant to the vertical force. The experimental result of vertical force shows an accuracy error of 0.249 kgf, a hysteresis error of 1.071 kgf, and a repeatability error of 0.202 kgf.^[100]

Another solution is to apply the prior knowledge to obtain multi-axial force from the known pressure. D. Jacobs et al.^[101] placed eight custom neoprene air bladders (SSCDANN030PGAA5 with area of 2.54 × 2.54 cm² from Honeywell well, Inc. USA) in shoes and trained a neural network with the detected pressure results from walking in an instrumented split-belt treadmill (Bertec Inc. USA) as shown in Figure 7b. After training the neural network, the plantar pressure in three directions can be extracted from raw pressure. When compared to a force plate, the normalized RMSE of the measured results is less than 15%.

In air pressure-based techniques, a sensing medium-fluid is needed to fill the sensors' cavity. Hence the sensor size is larger than its counterparts. In this case, the total ground reaction force can be more accurately detected. Nevertheless, its high volume and the large deformation under body weight can cause the user some discomfort and an unnatural gait. Hence, in practical applications, the sensor is more suitable for total plantar force-sensing cared diseases, for example, PD, but not for local pressure cared diseases, such as a diabetic foot.

2.1.7. Comparison of Plantar Stress Sensing Techniques

The advantages and disadvantages of the six sensing mechanisms are summarized in Table 1. It can be seen that the technologies based on piezoresistive techniques are less expensive

and commercially available, but suffer from considerable hysteresis and creep. Piezoelectric sensors are with better comfort and are favorable for massive production due to their thin-film structures. Optical methods are less sensitive to humidity, temperature, and electromagnetic interference; however, their suitable sensor structures need further studies. Techniques that well support shear force measurement are piezoelectric, inductive, and optical based architectures, among which the inductive-based methods are less used due to their high volumes.

In conclusion, the sensor material used for plantar pressure detection should meet comfort and high-precision acquisition requirements. The selection of sensing materials, electrode materials, and substrate materials with appropriate mechanical properties and sensing properties is the key to ensure the performance of the insole sensing system.

Table 1. Relative merits of plantar pressure sensing techniques.

Mechanism	Key materials	Upper detection limit	Lower detection limit	Sensitivity	Hysteresis error	Temperature drift	Advantages	Disadvantages	Appropriate diseases
Piezoresistive	Carbon ink, silver paste, conductive fabric, EGaIn liquid metal, MWCNT, PDMS, graphene, carbon nanotubes	56 kPa to 5 MPa	5 Pa to 30 kPa	11 MPa ⁻¹ to 15.1 kPa ⁻¹	7–50%	0.1–2.3 kPa/K	Low cost, simple structure and fabrication process, immune to EMI, support multi-axis pressure sensing	High hysteresis and creep, large temperature drift and power consumption	Support general chronic applications of neural diseases and orthopedic diseases
Capacitive	PDMS, MWCNT, carbon nanotubes	20–1200 kPa	3 Pa to 15 kPa	1.9 MPa ⁻¹ to 8.4 kPa ⁻¹	7–35%	1.1 Pa/K to 3.13 kPa/K	Low-cost, simple structure and fabrication process, low power consumption	Sensitive to temperature, humidity, and EMI noise	Support general chronic applications of neural diseases and orthopedic diseases
Piezoelectric	PZT ceramic, PZT powder, PDMS, FD-BTO	94.18–625 kPa	8.71 Pa to 0.7 kPa	2.7–4.4 pC/kPa	1–5%	9.4–62.5 Pa/K	Ruggedness, simple structure, low power consumption, high force sensitivity, suitable for dynamic applications, support multi-axis pressure sensing	High-cost, readout circuit sensitive to EMI noise, cannot detect static pressure	Support general chronic applications of neural diseases and orthopedic diseases, but cannot detect diseases which require static pressure analysis like RA
Inductive	Aluminum foil, mild steel, eGaIn, Ga wires, In wires	20–300 kPa	1.76–10 kPa	64–152 MPa ⁻¹	6.5–20%	Not given	Support multi-axis pressure sensing	High volume, complicated structure, sensitive to EMI	Suitable for the total plantar force-cared diseases like PD due to its large volume
Optical	GaAs, CdS, InP, ZnS for laser diodes, As ₂ S ₃ , As ₂ Se ₃ , PMMA, PDMS, PAM hydrogels for transmission module; CdS, CdSe, PbSe, PbS, CdTe for receiving end	200–1892 kPa	4 Pa to 10 kPa	0.2–26.4 MPa ⁻¹	2.3–20%	15.9 Pa/K to 5.2 kPa/K	Immune to EMI, low hysteresis and creep, support multi-axis pressure sensing	Require complicated measurement electronics and data acquisition system	Suitable for the local plantar force-cared diseases analysis like diabetic foot due to the complicated structure
Atmospheric pressure-based	Silicone, nylon, teflon, PU	125 kPa to 2.5 MPa	0.1–30 kPa	0.5 MPa ⁻¹	1–16%	0.5–1 Pa/K	Immune to EMI, suitable for whole insole pressure detection	High volume, and creep, difficult in integration	Suitable for the total plantar force-cared diseases like PD

2.2. Layout of the Sensing Elements

Although it is widely recognized that different sensor designs can have diverse effects, there is no unified agreement on the most optimized sensor design, such as the position and number, for specific disease or motion parameter detection. In this subsection, the deployment considerations of sensor size and sensor layout and their influence on the system capabilities are discussed to provide general guidance for future designs.

2.2.1. Sensor Size

The primary design consideration is the sensor size, which is directly related to the quality of raw data. A small sensor may easily miss the peak pressure position, especially when a relative slide occurs between the foot and the shoe, while a large sensor area leads to an underestimation of peak pressure, especially for the acute stress under the metatarsal bones and toes (e.g., a round-shaped electrode with a diameter of 10 mm can cause a reduction of 177–70% in peak pressure prediction, due to the averaging effect^[17,102–104]).

To obtain reliable plantar pressure information, M. Razian et al.^[105] recommend that the sensor size be no more than 10 mm × 10 mm. B. Davis et al.^[104] claim that sensor size larger than 6.36 mm × 6.18 mm may lead to sub-optimal sampling through experiments on diabetic patients and frequency-domain analysis. S. Urry^[17] suggests the size should not exceed 5 mm × 5 mm. V. Berki et al.^[106] found that a sensor size of 4.8 mm × 4.8 mm is appropriate for both normal and shear stress sensing. C. Giacomozzi^[107] recommends a diameter of 2–3 mm for the sensors under localized peak pressure region, such as the metatarsal heads area. For general purposes, the recommended sensor size for plantar pressure measurement is 5 mm × 5 mm,^[15,17,102] the accuracy of which has been proved to be more than 90% for maximum pressure estimation under metatarsal bones.^[108] A size smaller than that should be arranged as a sensor array to avoid missing critical details.^[15]

2.2.2. Sensor Layout

After determining sensing mechanisms and sensor size, the factors to further consider are the number, position, and arrangement of sensing elements. A conventional strategy is to place discrete sensors at preset locations. The method includes three steps: First, based on the foot structure or function, the plantar area is divided into several regions. Second, specific areas are selected according to the characteristics of the target disease. Third, within the chosen regions, the positions that are mostly associated with disease development are determined, such as the peak stress positions or symmetrical positions at the medial and lateral foot.

The most widely used way for the plantar region division relies on the anatomical position of bones and tissues,^[109] which split the insole area into 15 areas: 1–3 at the heel, 4–5 at midfoot, 6–10 at the forefoot, and 11–15 at the toes.^[110] These areas carry most of the body's weight and can reflect the lower limb balance. In **Figure 8**, the points labeled as 1–15 shows

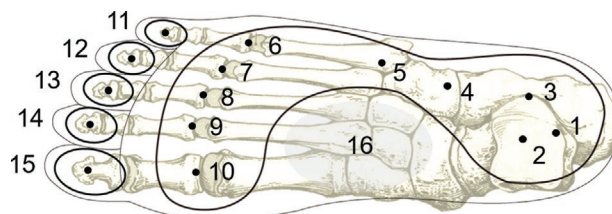


Figure 8. Plantar region classification according to the anatomical position of bones and tissues. Points with label 1–15 refer to the peak pressure positions and 16 refers to the arch area. Reproduced with permission.^[110] Copyright 2010, IEEE.

the key positions in each area, which are under the peak pressure points or correspond to some important anatomical structures^[109]; the region 16 is newly added to present the arch of the foot, which is useful in some specific applications, such as the diagnosis of flatfoot and clubfoot. Then, for each disease, the target plantar areas are selected based on its clinical symptoms of gait. For example, the foot ulcers of diabetes are more likely to occur at the positions with higher normal or shear stress, especially the hallux, forefoot, and heel.^[41,94,111,112] In ref. [112], eight sensors are distributed under the hallux, metatarsal bones, and heel, as shown in **Figure 9a**. The risk of ulcers can be determined by whether the normal stress values exceeding some predetermined thresholds. The gait analysis for stroke hemiplegia patients focuses on the symmetry of pressure and time parameters of two feet, rather than the absolute pressure values at specific points.^[51,113,114] Therefore, the sensing points are generally scattered at various high-pressure locations to ensure a high force response and a more comprehensive spatial analysis. For example, C. Wang et al.^[51] set ten sensors at the first and third toes, metatarsal bones, and the heel (**Figure 9b**) to examine patients' walking speed and gait asymmetries, and successfully distinguished the hemiparetic gait patterns from normal gait. For the arch structure, deformation-induced symptoms like flat foot and high arch, the ratio of medial to lateral foot pressure is of significance.^[115,116] As shown in **Figure 9c**, five sensors are symmetrically located at the medial and lateral foot for flatfoot detection.^[115] A significant difference in the pressure ratio is experimentally detected between flatfoot and normal foot.

In some cases, more sensor points should be set to improve the spatial resolution. For example, the diabetes foot ulcers may occur at a variety of locations on the plantar contact surface^[75] so that the targeted plantar area cannot be determined in advance; when designing orthotic insoles for diabetes and amputees, it is essential to observe the pressure distribution on the contact surface between the foot and insole.^[75] In such scenarios, a widely-used strategy is to distribute the sensors uniformly on the entire insole and reconstruct the pressure distribution with algorithms for subsequent analysis. In the sensor array arrangement, the spatial resolution and time resolution are both related to the detecting accuracy. A small sensor size allows high spatial resolution, hence improving the accuracy of plantar force reconstruction; simultaneously, the increasing component and data management cost limit the system scanning frequency, which may result in signal distortion after sampling. The fundamental requirement for desired detection accuracy is that

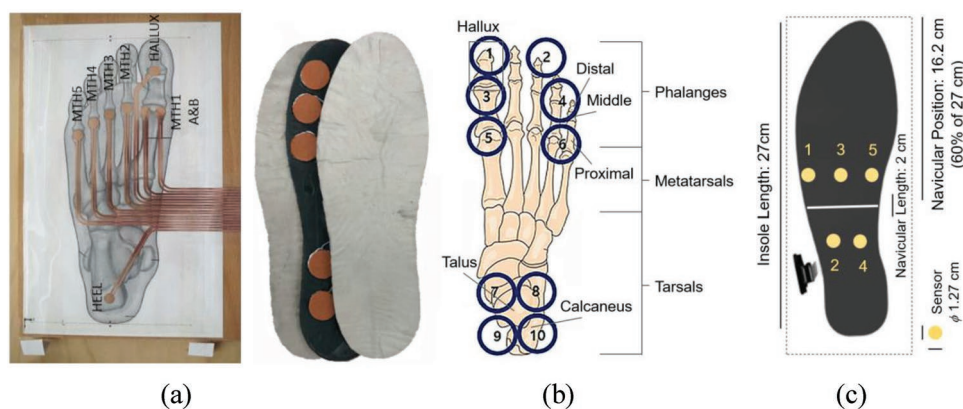


Figure 9. The insoles sensor layouts designed for the analysis of a) diabetic foot ulcers, b) stroke hemiplegia, and c) flatfoot. Reproduced with permission.^[51] Copyright 2019, MDPI. Reproduced with permission.^[113] Copyright 2016, MDPI. Reproduced with permission.^[115] Copyright 2018, Sensors.

both the temporal and spatial sampling frequency should meet the Nyquist-Shannon sampling theorem.^[102] It is generally considered that a 50–100 Hz scanning frequency is sufficient for daily activities.^[75,117] Work in ref. [118] revealed that a sampling rate of 25 Hz could guarantee good reconstruction quality when the walking speed is below 2 km h⁻¹. In some extreme cases, to analyze the impact force in some fast-moving actions, a scanning frequency of 300 Hz is required.^[119]

Furthermore, in recent years, downsampling techniques are applied to optimize the layout of insole sensor arrays. Researchers use insole systems with high spatial resolutions to conduct contrast experiments, find the highly associated regions with specific tasks, and then determine the optimal sensor location and combination mode. In 2014, K. Kanithika et al.^[120] found out the key areas on gait analysis by calculating the Pearson's correlation between 11 healthy subjects and 99 sensor position. Results show that the high correlation points are mainly distributed across four regions: toes, metatarsals, outline of barefoot, and heel (the positions are presented

in **Figure 10a**), providing important shreds of evidence for the post-downsampling-based sensor arrays design. With a similar controlled trial, E. Martini et al.^[121] select 16 sensors that reasonably reflect the gait phase from a 64-sensor array by visual inspection (shown in **Figure 10b**). The simplified system achieves an estimation accuracy of 98.7%, with only 25% of the measuring points. In ref. [122], points of an 8-sensor insole system are removed in turn to examine their relative importance in accessing the gait asymmetries of stroke patients. Four points are finally selected that are able to reflect the plantar pressure patterns for stroke patients optimally. The layout optimization of pressure sensors helps the system maintain high detection accuracy with reduced hardware costs and data processing time. Nevertheless, studies on this aspect of disease detection are rare and require devoting much effort.

Table 2 summarizes the requirements of plantar pressure sensors for different diseases, including target plantar areas, sensing mechanism candidates, pressure information in interest, and the size and layout of sensing points. It is worthy of mentioning that, for diabetic foot ulcers, the concerned information is the absolute pressure magnitude, while the rests focus on the relative quantities, such as symmetry and the ratio of pressure at each region.

In the past decades, numerous insole pressure sensing sensors for various disease detection applications have been proposed in academic researches and continuously emerged in commercial products. The sensors are continually developing in the direction of miniaturization, specialization, high accuracy, and massive production. This section overviews the sensing mechanisms and sensor layouts of the insole systems in the literature. Each detection mechanism has its distinct pros and cons that determine sensor performance. Among them, the piezoresistive and capacitive methods are widely used in normal stress detection; the measurement of shear force is mostly achieved by measuring the shear strain, where piezoelectricity and optical fiber are most suitable for insole sensing. The sensor location is usually determined according to the pressure center and can be further optimized through correlation analysis methods. The recommended size of the detection point is around 5 mm × 5 mm. When the interest region is uncertain, the array sensor is usually used to measure

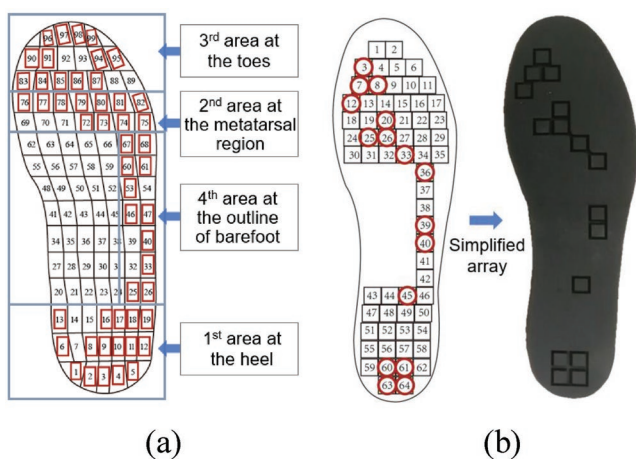


Figure 10. a) The key areas (circled) for gait analysis selected by correlation coefficients in the study of ref. [120]. Reproduced with permission.^[120] Copyright 2014, IEEE. b) The 16 points (circled) selected from a 64-sensor array and the simplified insole device. Reproduced with permission.^[122] Copyright 2017, IEEE.

Table 2. Summary of sensor parameters and interested plantar areas for diverse diseases.

Disease	Plantar regions of interest	Available sensing mechanisms	Concerned pressure information	Sensor number (for discrete sensors)	Spatial resolution ^{a)} (for array)	Sensor side length [mm]
Diabetic foot ulcers ^[34,35,46,67,72,98,99]	Hallux, forefoot, midfoot, heel	Piezoresistive, piezoelectric, capacitive, inductive, optical	Peak values, upper sensing limit should be >450 kPa	5–8	>0.14	<15
Parkinson disease ^[24,40,50,100,123]	Hallux, forefoot, arch, midfoot, heel	Piezoresistive, piezoelectric, capacitive, air pressure, optical	Ratio of pressures at interest regions, peak pressure frequency	4–16	>0.27	<40
Stroke hemiplegia ^[74,224,225,228]	Toes, forefoot, midfoot, heel	Piezoresistive, piezoelectric, capacitive, optical	Pressure distribution similarity of two feet	4–8	Not suitable	<12.7
Knee osteoarthritis ^[21,226,227]	Toes, forefoot, midfoot, heel	Piezoresistive, piezoelectric, capacitive, optical	Ratio of pressures at interest regions	4–8	Not suitable	<12.7
Elder falling event ^[52,53,68]	Toes, forefoot, midfoot, heel	Piezoresistive, piezoelectric, capacitive	Pressure distribution pattern abnormality	6–14	>0.27	<12.7
Flat foot ^[39,223]	Forefoot, midfoot, arch, heel	Piezoresistive, capacitive, optical	Ratio of pressures at interest regions	≥3	Not suitable	<50

^{a)}The spatial resolution is calculated as the average sensor number in 1 cm² area.

the pressure distribution. Finally, the review of sensor setup in different disease diagnosis may provide a reference for future design work.

3. The Electronics Design for Pressure Sensing

After fabricating the insole sensors for specific disease monitoring, the electronic analog front ends are constructed. The analog front ends are responsible for converting the sensing information to a readable electrical signal (voltage, current, or even digital signal). For different sensing mechanisms, corresponding analog front ends are distinct. This section will provide the readers with the broadly utilized circuit designs for different pressure sensing mechanisms.

3.1. Resistive Techniques

The circuits used for resistance detection can be divided into two categories due to the structures: voltage divider and Wheatstone bridge. The voltage divider circuit can be implemented to detect the resistive sensors, which have the resistance of several million ohms with no load and tens of kilo-ohms with a full load. The sensor's output impedance is high (>10 kΩ), generating unnecessary voltage drop due to the input current of the following analog-to-digital conversion, hence reducing the measurement accuracy. To address this, a widely utilized analog front end architecture is presented in **Figure 11**. A resistive sensor R_x and a precision reference resistor R_{ref} are connected in series between the power supply terminal (V_{cc}) and the ground. The current driving capability is amplified with a voltage follower, and the output voltage V_{out} is presented in Equation (1):

$$V_{out} = \frac{R_f}{R_f + R_x} \times V_{cc} \quad (1)$$

As to the former circuit type, the Wheatstone bridge circuit occupies one, two, or four sensing bridge arms depending on the application scenarios. **Figure 12** shows a widely used framework with the sensing resistor R_{g1} and R_{g2} . With no load,

the bridge is set to a balanced bridge with the relationship as shown in Equation (2).

$$\frac{R_1}{R_{g1}} = \frac{R_2}{R_{g2}} \quad (2)$$

In this case, there is no voltage difference between points B and D (output voltage: $V_{bridge} = 0$ V) when no load is being applied. However, when the pressure sensor suffers from the external pressure, the output voltage (V_{bridge}) of the Wheatstone bridge is equivalent to the voltage difference across the R_{g1} and R_{g2} following Equation (3):

$$V_{bridge} = V_{cc} \cdot \left[\frac{R_{g1}}{R_1 + R_{g1}} - \frac{R_{g2}}{R_1 + R_{g2}} \right] \quad (3)$$

Equation (3) is advanced with no current passing through the voltmeter (to detect V_{bridge}) from the points B and D. Besides, the resistance change rate ($\Delta R/R$) of this technique with full sensing scale is about 1%, resulting in the mV-level change of

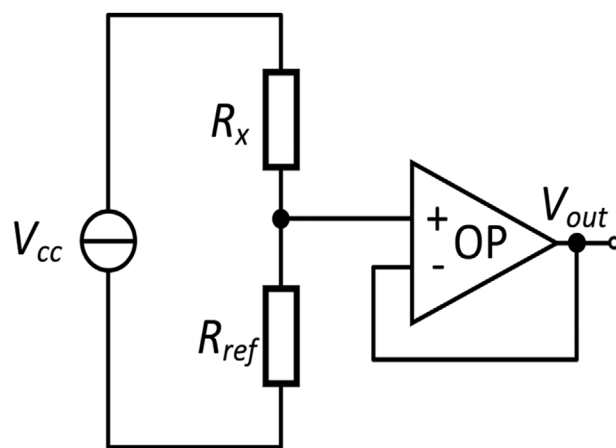


Figure 11. A block diagram of carrier mobility-based pressure sensor analog front end.

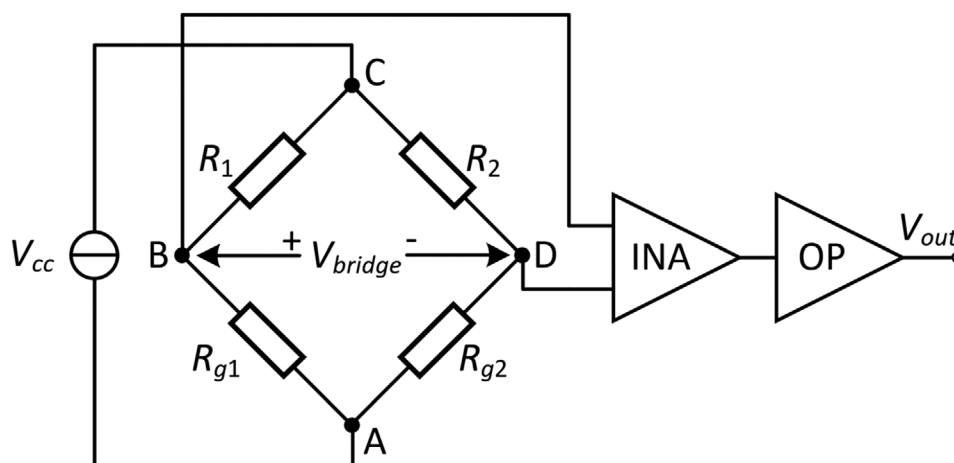


Figure 12. A block diagram of element geometry/structure change-based pressure sensor analog front end.

the output voltage V_{out} . In this case, the analog front end needs to represent a high input impedance to the sensing element and provide a high and stable gain for a small voltage signal.

Given the above requirements, multi-stage amplification is generally used in the analog front end (Figure 12). First, the instrument amplifier (which provides low DC offset and high input impedance) magnifies the slight voltage difference between points B and D without distortion. Second, the output voltage of the instrument amplifier is further amplified through the proportional amplification circuit to match the input range of the following analog-to-digital conversion circuit. Under these settings, assuming that the sensing signal output by the bridge is V_{bridge} and the total gain of the signal conditioning circuit is G , the final output, V_{out} , of the analog front end is shown as shown in Equation 4:

$$V_{out} = G \times V_{bridge} \quad (4)$$

3.2. Capacitive Techniques

The widely utilized capacitance detection schemes can be divided into RC time constant schemes, LC resonator (also named as LC tank) scheme, and capacitive voltage divider. In what follows, we will briefly describe these three frameworks.

In the scheme of the RC time constant detection, a fixed resistor is connected to the sensing capacitor. The capacitance value can be determined by detecting the circuit time constant RC as shown in Figure 13. The sensing cycle is to charge the capacitor first with I_{charge} and then record the voltage dropping

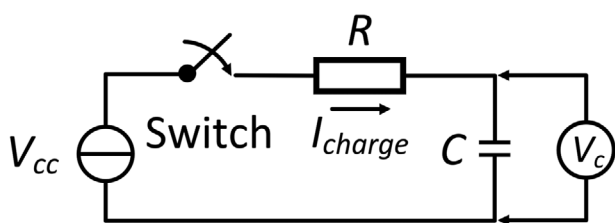


Figure 13. A block diagram of RC time constant-based capacitance sensing circuit.

time until the capacitor voltage (V_c) drops to the set value by using the analog-to-digital converter^[124–126] or timer.^[127] Once the time constant has been calculated, solve for the capacitance value. Though the RC time constant-based circuit architecture is relatively simple, the scheme is susceptible to EMI noise due to the broadband sensitivity.

The LC tank capacitance detection architecture is shown in Figure 14, which was first proposed by Texas Instruments and applied to the FDC2 × 1× product.^[128] A resonant circuit driver is integrated into the chip and followed by a multiplexer. One end of the sensing capacitor C_{sensor} (take $C_{sensor1}$ as an example) is connected to an external LC resonator, and the other end is grounded. In the working cycle, the resonant circuit driver will first activate the external LC resonator circuit and make it oscillate stably; then the chip core will compare the frequency of the clock source (internal clock or external clock with the frequency of f_{ref}) and LC oscillation circuit, and output the frequency ratio DATA (digital stream). The resonant frequency $f_{sensor1}$ of the LC oscillation circuit can be determined according to Equation (5):

$$f_{sensor1} = \frac{CH_FIN_SEL \times f_{ref} \cdot DATA}{2^{28}} \quad (5)$$

in which CH_FIN_SEL is the resonant frequency division coefficient of the external LC oscillator circuit. The sensing capacitance $C_{sensor1}$ is given in Equation (6):

$$C_{sensor1} = \frac{1}{L_1 \cdot (2\pi \cdot f_{sensor1})^2} - C_1 \quad (6)$$

in which L_1 is the inductance of LC tank resonator, and C_1 is the total capacitance parallel with L_1 . Compared with the RC time constant-based detection scheme, this narrow-band sensing method provides a very high anti-EMI rate and significantly increases the signal-to-noise ratio.

The transmission technique utilizes the phenomenon in which the AC current passing through the capacitor is

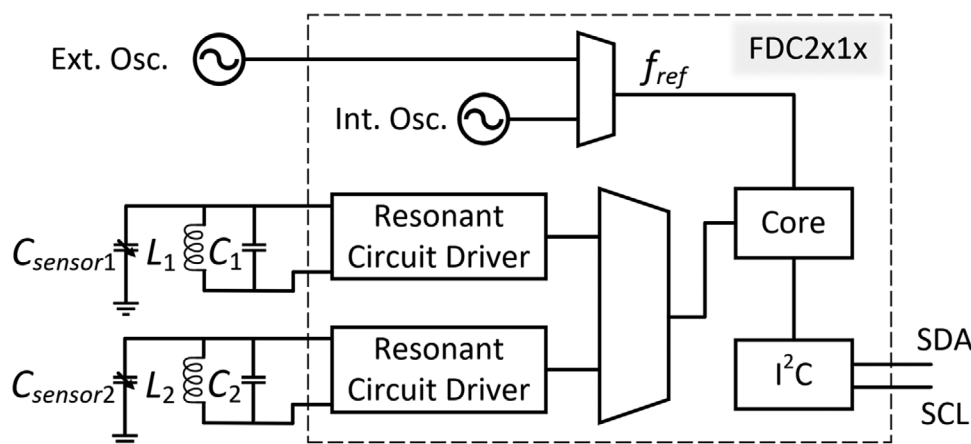


Figure 14. A block diagram of LC tank capacitance detection architecture.

negatively related the capacitance value, and the capacitor voltage division scheme (shown in Figure 15) is the representative approach. The sensing capacitance is connected to the excitation voltage source (sinusoidal or rectangular wave signal with a specific driving frequency), and determines the current that passes through the variable capacitance. With the charge amplifier, the AC current is then converted into a voltage, V_{charge} , with a feedback capacitance C_f as shown in Equation (7):

$$V_{\text{charge}} = \frac{1}{C_f} \int I \cdot dt \quad (7)$$

V_{charge} then enters the rectifier circuit and low-pass filter where it outputs a DC signal, V_{out} , proportional to $C_x/(C_x + C_f)$ as shown in Equation (8).

$$V_{\text{out}} \approx \frac{C_x}{C_x + C_f} \quad (8)$$

In this case, the capacitance value of the sensing capacitor C_x can be determined. Besides, EMI noise can be suppressed by adjusting the excitation frequency of the voltage source due to the narrowband detection scheme.

Among these sensing frameworks, the RC time constant scheme provides a simple circuit structure for capacitance

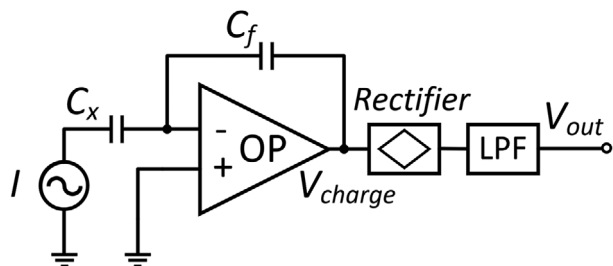


Figure 15. A block diagram of capacitor voltage division scheme.

sensing, which is suitable for a low EMI environment. LC tank-based and capacitive voltage divider-based circuits produce a high EMI rejection rate, supporting plantar pressure detection in most instances. However, the comprehensive circuit architecture increases the circuit cost and circuit volume, which is not feasible in practical applications.

3.3. Piezoelectric Techniques

When applying the dynamic force to the piezoelectric sensor, free charge, Q , will be generated on the two polar plates. At the same time, a voltage difference, V_{piezo} , will be generated between the leads with Equation (9):

$$V_{\text{piezo}} = \frac{Q}{C_{\text{piezo}}} \quad (9)$$

in which C_{piezo} is the intrinsic capacitance of the piezoelectric sensor. Based on this, two typical schemes (proportional amplifier and charge amplifier) are utilized to amplify the voltage or charge generated from piezoelectric sensors.

In the proportional amplifier scheme (as shown in Figure 16a), one end of the piezoelectric sensor is connected to the circuit ground, and the other end is connected to the positive input of the proportional amplifier. Hence the total capacitance paralleled to the piezoelectric sensor is C_{piezo} and amplifier input capacitance C_{in} . The amplification coefficient, G , is determined by the resistances R_L and R_m , and the final output V_{out} is:

$$V_{\text{out}} = \frac{Q}{C_{\text{piezo}} + C_{\text{in}}} \times \frac{R_L + R_m}{R_m} = \frac{Q}{C_{\text{piezo}} + C_{\text{in}}} \times G \quad (10)$$

The environmental variation (e.g., humidity) will affect the C_{in} and modify the amplification coefficient G .^[129] Besides, the applied pressure will change the distance between the two polar plates of the piezoelectric sensor, resulting in the instability of C_{piezo} . Therefore, the circuit architecture based on voltage amplification is not suitable for high-precision detection.

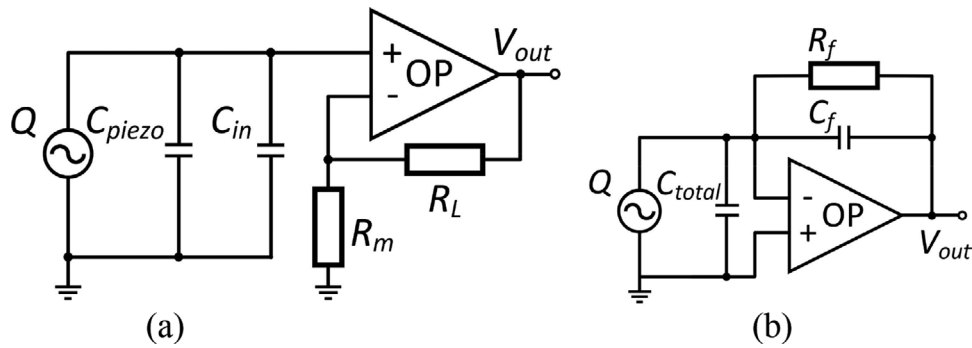


Figure 16. Block diagrams of analog front end utilized in piezoelectric sensors: a) proportional amplifier and b) charge amplifier.

The charge amplifier architecture is shown in Figure 16b. Two ends of the sensor are respectively connected to the amplifier's positive input and negative input, and the feedback resistor, R_f , alongside the feedback capacitor, C_f , determine the cut-off frequency, f_L , as shown in Equation (11):

$$f_L = \frac{1}{2\pi R_f C_f} \quad (11)$$

When piezoelectric sensors suffer the external pressure, the force-induced charges, Q , accumulate on the feedback capacitor, C_f , and produce the output voltage, V_{out} . Above the cut-off frequency, f_L , the output voltage is proportional to Q as shown in Equation (12):

$$V_{out} = -\frac{Q}{C_f + C_{total}/A} = -\frac{Q}{C_f} \quad (12)$$

in which C_{total} is the total input capacitance with amplifier input capacitance, C_{in} , and sensor's intrinsic capacitance, C_{piezo} . The amplifier open-loop gain A is extremely large (usually $> 10\,000$), making C_{total} effect negligible. Through the appropriate selection of R_f and C_f to set the desired cut-off frequency for insole plantar pressure detection,^[130] this method can detect plantar pressure with high accuracy and stability.^[19,48]

3.4. Inductive Techniques

For inductance detection, both using a coupling effect and building an oscillator circuit are feasible. For the former,

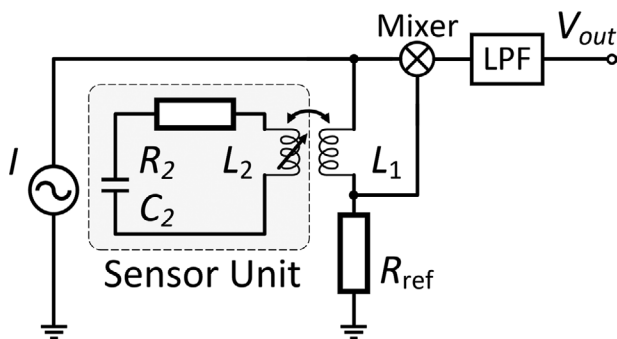


Figure 17. A block diagrams of near-field electromagnetic field coupling for inductive sensor detection.^[131]

refs. [85,131] report a wireless detection method for the inductive sensor using the near-field electromagnetic field coupling effect, as shown in Figure 17. A current source generates a sinusoidal current I with adjustable frequencies, and the current I passes through the reference inductor (receiving coil, L_1 , coupled with sensing inductance L_2) and reference resistor R_{ref} . The mixer sampled the voltage difference across L_1 , and with a low-pass filter, the output voltage V_{out} is associated with the phase drop due to L_2 coupling. According to the electromagnetic field coupling theory, the maximum frequency point of phase loss, $f_{phase-min}$, is related to the intrinsic resonant frequency (f_s) of the sensor as shown in Equation (13):

$$f_{phase-min} \approx \left(1 + \frac{1}{4}k^2 + \frac{1}{8Q^2}\right)f_s \quad (13)$$

in which k is the coupling coefficient between sensor antenna and receiving coil, and Q represents the quality factor of the sensor. Hence, the measured $f_{phase-min}$ is proportional to f_s when the two factors are constants. The resonant frequency, f_s , is only determined by the circuit itself and can determine the value of inductance, which in turn can be used to obtain the pressure data. This method can realize wireless pressure detection, and the inherent narrow-band sensing characteristics can effectively suppress the EMI from other spectral bands. However, scanning the wideband with the high-frequency resolution is required to determine $f_{phase-min}$, reducing the sampling process. Besides, the distance between sensor antenna and receiver coil will extremely affect the coupling coefficient, k , making it difficult for fast measurement.

As to the latter, ref. [132] presents another resonant frequency-based method to detect the inductance. As shown in Figure 18, three RLC resonators with different resonant frequencies are connected in series, and the excitation signal composed of three sine waves (whose frequency is set to the resonant frequency without load) is applied at the same time. Since the signal from each sensing inductor has a peak amplitude at its resonant frequency, the inductance variation of each circuit can be recovered from the combined response by obtaining the amplitude at each resonant frequency.^[133] With a high-resolution spectrum analysis (e.g., implemented by digital signal processor), amplitudes at each resonant frequency can be determined simultaneously, which can be utilized in plantar pressure detection with multiple channel sensing elements.

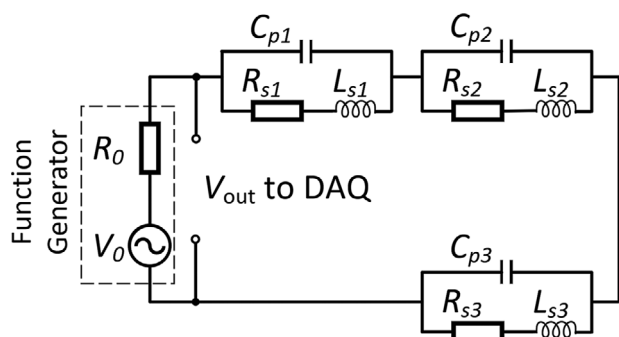


Figure 18. A block diagrams of three resonating RLC circuits for inductive sensor detection.^[132]

3.5. Optical Techniques

Optical detection schemes can be divided into fiber- and no fiber-based methods. Both sensing categories utilize photodiodes to detect the photocurrent as shown in **Figure 19**. The cathode of the photodiode is connected to the negative input of the amplifier, and the anode is connected to the negative power supply ($-V_{cc}$), making the photodiode work in the reverse bias region. When the light intensity increases, the photocurrent, I_{Light} , which flows through the photodiode also increases. The output voltage, V_{out} , is generated with the feedback resistor, R_f , as shown in Equation (14).

$$V_{\text{out}} = R_f \times I_{\text{Light}} \quad (14)$$

The light intensity change can be determined by reading V_{out} . Compared with the former scheme, the power consumption is extremely reduced, and complex post-processing optical path and circuit are not required. Hence, this scheme can be easily integrated into the wearable plantar pressure detection system.

3.6. Air Pressure Based Techniques

The typical structure of air pressure-based sensors contains air chambers (e.g., winded tubes or elastomeric air pillars) and a

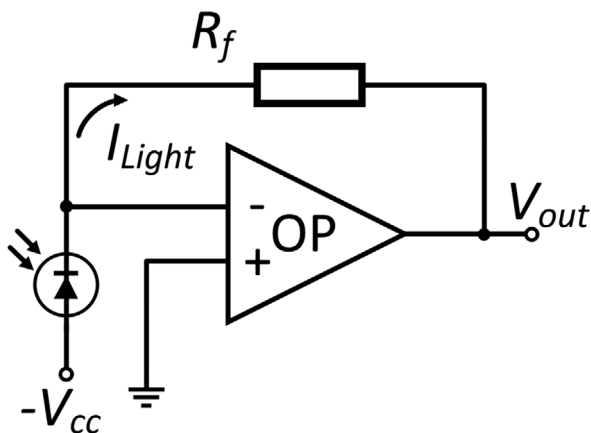


Figure 19. Block diagrams of photocurrent detection circuit.

pressure sensor for air pressure detection. Based on utilized sensing mechanisms, the air pressure sensor can be divided into resistance type (mostly the structure of strain gauge),^[134] capacitance type,^[135] and inductance type^[136] based on the sensing architecture and the associated circuits discussed in the previous chapters. Besides, there are many commercial integration schemes on the market, which can directly convert the collected data into digital signals. These include Bosch's barometric pressure sensors-BMP280 (resistive technique)^[134] and Hefei WNK Smart Technology Co., Ltd., W21 (capacitive technique),^[135] which can be easily integrated into the wearable plantar pressure detection system.

To sum up, choosing the appropriate circuit architecture based on different applications and electrical characteristics of pressure sensors is the key to obtain high-precision sensing data for further data processing. This section may provide guidance for designing a high-performance circuit for plantar pressure detection.

4. The Algorithm of Plantar Pressure Distribution

The previous section discusses important aspects of designing the insole front end for diverse application scenarios. However, raw data cannot be directly used for disease analysis. Hence, algorithms are developed to extract identifiable parameters of interest, that is, CoP, contact time, and cadence, directly linked to the development of diseases.

In the past twenty years, fruitful algorithms have been presented in the literature to address this issue. In this section, we first discuss algorithms for obtaining the plantar pressure distribution with high resolution, which will provide necessary information for disease-related features extraction. Next, algorithms for plantar pressure mapping are evaluated and compared to different subjects with various foot sizes and foot orientations. Finally, the algorithms for extracting parameters of interest from plantar pressure are articulated.

4.1. Reconstruction of Plantar Pressure Distribution

Generally speaking, to obtain high plantar pressure resolution, according to the Nyquist theorem, a high spatial frequency is required, which means that a high density of sensing points should be utilized and sampled at the detection process. However, it is challenging to install enough plantar sensors for wearable sensing systems due to the limited space. Researchers have proposed various reconstruction methods to solve this problem using the fitting, compressed sensing (CS), and machine learning-based algorithms.

4.1.1. Fitting

Fitting-based reconstruction methods (e.g., interpolation) are usually used to improve the data distribution resolution obtained from sensors. The supplemented data are generated from truly captured pressure signals, increasing the resolution.^[137] It has been successfully used in FPs and some

commercial instrumented shoes (e.g., Tekscan tactile sensor^[42]) with a high density of sensors to further improve the spatial resolution. However, some studies^[104,138] claim that fitting methods are not appropriate when the density of the sensor is limited (less than 60 sensing points at one foot fails to satisfy the Nyquist theorem).

In 2015, A. Tan et al. used a 75 sensing nodes based insole system with varying degrees of interpolation to effectively improve the gradual progress of visualizing the plantar pressure map from the raw data. Compared with the Kistler force plate, the results after interpolation have a residual standard deviation (SD) of 70.35 N when the accurate force is higher than 700 N.^[139] In 2019, 16 electrically segmented pressure sensors are fabricated by W. Lee et al. in shoes, and an interpolation scheme is used to determine the 16 pressure data as the source pixels as Equation (15):

$$U_{ns}(i, j) = N \times \left[\sum_{s=0}^{s=15} U_s \times e^{-k \times d_{ijs}} \right] \quad (15)$$

in which $U_{ns}(i, j)$ is the value of no source pixel, N is the normalizing factor, U_s is the value of the source pixel, k is the index, and d_{ijs} is the distance between (i, j) -th no source pixel and s -th source pixel. After reconstruction, the foot outline was determined, but the detail of the foot structure could not be distinguished.^[140]

In summary, the limited sensor density of the consumers' affordable wearable device brings significant challenges to the high spatial resolution reconstruction of plantar pressure patterns through the fitting method. Hence, it is not conducive to accurately determine the progress of related diseases.

4.1.2. Compressed Sensing

For fitting-based reconstruction techniques, the Nyquist theorem restricts the maximum reconstruction resolution to half of sensor placement density. Besides, current fitting methods take the same mapping relationship of whole feet rather than considering physiological differences, which further reduces the reconstruction accuracy.

Through studying the characteristics of plantar pressure distribution, researchers reveal that the mapping is sparse at

domains in which plantar pressure distribution can be represented as a linear combination of specific models.^[141] In this context, CS, an effective reconstruction method when the measured signal is sparse in a specific domain,^[142] is applied to gait signal reconstruction. With the advantage of small data volume and low spatial sampling rate, many researchers claim that CS has excellent potential in capturing and processing human motion signals.^[143] For insole plantar pressure reconstruction, CS has been successfully adopted with the desirable resolution.^[104,141,143]

The first step of utilizing CS is to find the sparse domain of the plantar pressure distribution, in order to select basic functions since its representation is quite significant. In 2013, S. Ostadabbas et al. presented a linear combination of several Gaussian functions to express the plantar pressure distribution. Then principal component analysis is used to reduce the number of free variables. To train the model, the high-resolution pressure data collected by the Tekscan FP is used to establish the regression and mapping relationship of each subject. The free variables can then be determined from the pressure information of five pressure sensors (arranged at the positions of big toe, heel, first to fifth metatarsal heads, one foot) with the least square method (LSQ) method. After that, the plantar pressure distribution condition is reconstructed, as shown in **Figure 20a** with a RMSE of 30 kPa.^[104]

The high-resolution pressure distribution is utilized to find a sparse representation of plantar pressure first.^[141] The researchers utilize the collected data from four sensors (arranged at the positions of thumb, heel, first metatarsal, and fourth/fifth metatarsal) with orthogonal matching pursuit (OMP) or least absolute shrinkage and selection operator (LASSO) algorithm to reconstruct the plantar pressure distribution. The LASSO shows half of RMSE compared to OMP, but the OMP is of low implementation and computation cost, and suitability for real-time application. As a result, the RMSE for each sensing point is within 6.7 kPa. When the pressure value is up to 160 kPa, the RMSE error is kept below 16 kPa.

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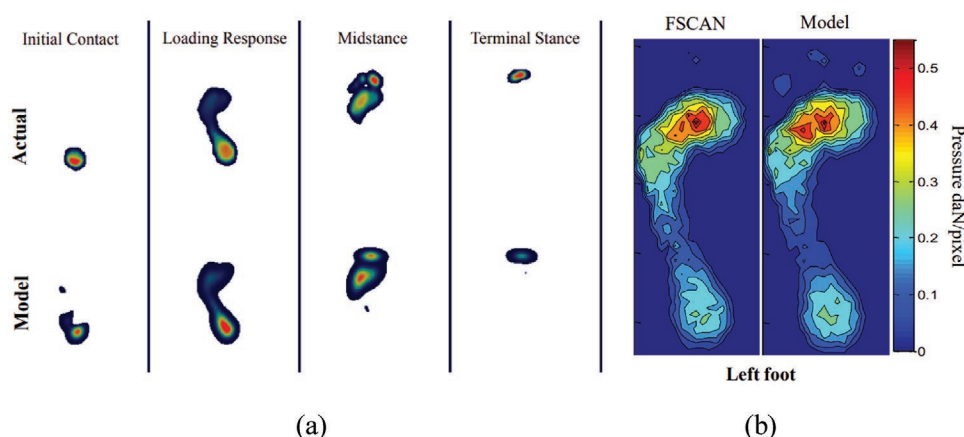


Figure 20. Reconstruction results of plantar pressure by a) CS, and b) machine learning. Reproduced with permission.^[104] Copyright 2014, IEEE. Reproduced with permission.^[145] Copyright 2014, Elsevier.

(LASSO) algorithm to reconstruct the plantar pressure distribution. The LASSO shows half of RMSE compared to OMP, but the OMP is of low implementation and computation cost, and suitability for real-time application. As a result, the RMSE for each sensing point is within 6.7 kPa. When the pressure value is up to 160 kPa, the RMSE error is kept below 16 kPa.

4.1.3. Machine Learning

Machine learning is a widely utilized method to allow the systems to learn from experience automatically, enabling disease symptoms to be classified accurately.^[15] Therefore, machine learning allows a sensing system with low sensing density to reconstruct the high-resolution plantar pressure pattern map.^[145,146]

In 2014, H. Ghaida et al. divided the foot structure into two parts: foot skeleton and elastic medium, and assume that the geometric movement of the foot plane and foot skeleton is equivalent to the geometric change caused by elastic medium compression. Machine learning methods are utilized to determine the structure of the foot skeleton and elastic medium.^[145] The information about the deviation of the plantar plate and the foot skeleton of the plantar pressure is obtained with the collected data from the three FSR sensors (arranged at the positions of the heel, first metatarsal, and fourth metatarsal),^[146] and the pressure distribution is determined.^[145] In this study, the average error is 2% of full scale with 0.1 N average error of each sensor, and the SD is 0.05 N (as shown in Figure 20b).

The comparison of these three reconstruction methods (fitting, CS, and machine learning) is summarized in Table 3.

4.2. The Registration of Plantar Pressure Mapping

Table 3. The comparison of three full plantar region reconstruction methods.

Reconstruction method	Advantages	Disadvantages
Fitting ^[139,140]	1) No training needed 2) Easy to implement	1) The accuracy is limited when the sensor number is smaller than 60 ^[132] 2) Higher accuracy requires higher sensing density 3) High energy consumption 4) Do not have physiological explanation
CS ^[104,141,144]	1) Only several insole sensors are needed to reconstruct the high-resolution plantar pressure	1) Require periodic calibration 2) Mapping relations need to be solved first 3) Require the randomness of sampling process
Machine learning ^[145,146]	1) Outstanding ability to establish nonlinear relationship between parameters of interest 2) Only several insole sensors are needed to reconstruct the high-resolution plantar pressure	1) Rounding errors from calculation approximation 2) Require periodic calibration

After obtaining the plantar pressure with high spatial resolution, the comparison with the reference pathological plantar mapping will be a necessity for further pathological features extraction and specific disease diagnosis. However, both the foot size and foot orientation are individually dependent, indicating that wrong correspondence and diagnosis are highly possible. To solve this issue, some registration algorithms (e.g., moments of the area method,^[147] massunivariate approach,^[148] etc.) have been developed, by which plantar pressure mapping is aligned with uniform size and orientation, enabling more sophisticated analysis techniques like automatic disease diagnosis based on artificial intelligence (AI).^[149]

As early as 2000, A. Harrison and P. Hillard used the moments of the area method to align the map of the plantar pressure distribution with the calculated principal axes automatically. The central axis' estimated angle and the center position of the foot-print are within 1° and 0.6 pixel of the actual value, separately.^[147] In 2008, the pedobarographic statistical parametric mapping technique was presented to record plantar pressure images. It made the homology structures of different models' best overlap by T. Pataky et al. Compared with ten-region subsampling, the mass univariate approach achieves pixel-level statistical analysis and exhibited a more significant negative correlation of walking speed and lateral midfoot peak pressure ($p < 5 \times 10^{-5}$).^[148]

N. Keijsers et al. asked twelve volunteers to walk on the FP in a straight line and at an approaching angle of approximately 40°. They develop a rotation algorithm to determine the foot progression angle of the subjects' walking direction and combined with the foot size to standardize the plantar pressure pattern with the mean shortest distance between the contour lines of straight walking and waling in an angle of 0.22 cm (SD: 0.06 cm) for the forefoot and 0.14 cm (SD: 0.06 cm) for the heel in 2009.^[149] In the same year, two algorithms, the 'geometric' algorithm, which exclusively matches geometry, and the 'hybrid' algorithm, which performs subsequent pseudo-optimization, are developed by Oliveira. The results show that subsequent pseudo-optimization can better minimize the mean square error difference and improve the overlap rate ($p < 0.010$) as well as calculation efficiency (each pair of images can be matched within 6 s) with the RMSE of $1.22 \pm 1.10 \text{ N cm}^{-2}$.^[150] Later, in 2014, the same research group proposed two image registration methods based on Fourier transform, utilizing cross-correlation optimization to match different images in the frequency domain and distribution phase, and obtained the RMSE of $2.01 \pm 1.45 \text{ N cm}^{-2}$. This method is robust to moderate noise (Gaussian noise with the mean amplitude of zero and SD ranging from 0 to 1.5 N cm^{-2}), and the time consumption is less than 33 ms.^[151] In 2019, Xia et al. present a cascaded convolutional neural network (CNN) method for the plantar pressure images registration and achieve the RMSE of $1.81 \pm 1.15 \text{ N cm}^{-2}$ with the average running time of $54 \pm 7 \text{ ms}$.^[152,153]

4.3. Gait Parameters Extraction

With a high spatial resolution plantar pressure distribution, other momentous gait parameters like GRF and CoP, can be extracted for an inclusive acknowledgment and diagnosis of the disease.^[154,155] Using certain extraction methods, like numerical

calculation and machine learning, important gait parameters from raw plantar pressure data can be extracted.

4.3.1. Direct Means

Some gait parameters (step, stance, CoP, vertical GRF, etc.) can be calculated with numerical calculation (e.g., summarizing, weighted average, threshold division, etc.). Therefore, low calculation time consumption and simple hardware requirements are required for these features extraction, making them easy to implement in microcontrollers.

In 1998, B. Davis et al. calculated the full vertical GRF of forefoot based on the direct addition of the pressure value from a 4×4 sensor matrix with the simple summarization method. The method reveals excellent linearity ($\pm 5\%$), low hysteresis ($\pm 7.5\%$), and good agreement between actual and measured loads ($\pm 5\%$).^[156] In 2000, the Parotec commercial insole pressure sensing system (24 sensing points each foot) was placed between the two layers of socks worn by users to calculate the CoP in the x direction (medial-lateral direction) with the weighted average, as Equation (16):

$$\text{CoP}_x = \frac{\sum x_i P_i}{\sum P_i} \quad (16)$$

in which x_i is the pressure pixel location, P_i is the pressure value, and the denominator $\sum P_i$ demonstrates the vertical GRF. The CoP in the y direction (anterior-posterior direction) has a similar equation. In this study, the CoP origin was confirmed through wetting the outermost layer of socks. The RMSEs of x and y direction are 5.6 mm and 13.8 mm, respectively.^[157] In 2014, S. Crea et al. used a flexible insole with 64 pressure-sensitive elements to collect data at a sampling frequency of 100 Hz and transmitted it through Bluetooth. The parameters of vertical GRF and step cadence are calculated with a summarization method and zero detection. However, the estimated vertical GRF result showed a normalized RMSE of 106.1 N compared with FP.^[87] As the increment of sensor density, the RMSE drops. In 2019,^[158] the vertical GRF and CoP obtained from the insole system (88 sensing points each foot) were compared with the Kistler plate through a weighted average based technique. As a result, the RMSE of CoP of x direction and y direction is within 4 mm and 10 mm, respectively.

4.3.2. Calculation with Machine Learning

Although the numerical calculation method can extract some essential gait parameters for clinical diagnosis and rehabilitation, their non-linearity undercuts the accuracy of the estimated results; indirect methods are adopted to address these problems. Machine learning can be implemented for this kind of problem and is favored by researchers.

In 1999, H. Savelberg demonstrated a specific mapping relationship between plantar pressure distribution and shear GRF.^[159] Based on this prior study, in 2008, D. Fong et al. utilized the pressure data from Pedar sensing insole (99 sensing points) and combined it with the platform results to establish

the regression model. The model takes the readings of several selected channels as input and solves the total force at three directions, anterior-posterior, vertical and medial-lateral direction, respectively, and the maximum RMSE appears in the vertical direction with 38.43 N.^[160] In 2006, M. Chen et al. used eight FSR sensors data with the support vector machine (SVM) regression algorithm to solve the relationship between the FSR values and the Pedar insole system's readings. After that, the mean square error of the estimated result from FSR sensors is 0.87 kPa.^[154] In 2012, L. Lincoln et al. introduce a silicone instrumented shoe to measure the GRF in all three directions. The fitting regression method is based on five sensing arrays, five pulsed emitters and five detectors (125 total signals). The model of the form is shown in Equation (17):

$$\text{Force} = \sum \alpha_i D_i E_i \quad (17)$$

in which, α_i is a regression of i -th signal, $D_i E_i$ is the response from detector i to emitter i , and Force is the desired force. The ME of the final shear GRF is 10.7 N, and ME of the vertical GRF is 68.1 N, which is within 10% of the full-scale mistake.^[89] In 2017, P. Rosquist et al. used four novel nanocomposite piezo-responsive foam sensors and calibrate separate models for each individual by a tenfold cross-validation process. The model estimated average anterior-posterior, mediolateral, and vertical GRF with the mean absolute error (MAE) of 6.52 N (2.14%), 4.79 N (6.34%), and 15.4 N (2.15%), respectively. Besides, the MAE increases if the personalized model was not adopted.^[161] All these works of obtaining the gait parameters through direct and indirect calculation are summarized in Table 4.

5. Plantar Pressure-Centered Multisensory Fusion

Plantar pressure distribution and related extracted parameters obtained from pressure sensors can satisfy most of the clinical applications. However, for those requiring more detailed gait information (like energy expenditure monitoring for sports damage evaluation), merely utilizing plantar pressure distribution information would be not enough. In this situation, multisensory fusion-based techniques are proposed, in which multidimensional information is received through collaborations with other sensors, such as EMG sensors and inertial measurement units (IMU, including accelerometer and gyroscope). A scheme of insole gait monitoring system with IMU, EMG and FSR sensors is presented in Figure 21.

Here, we briefly explain why other dimensional information can provide additional important gait details by giving two examples. First, EMG information is essential for monitoring neuromuscular activities,^[162] and evaluating patients' muscle activation,^[163] and can serve as a prominent reference for muscle dysfunction treatment.^[164] Second, IMU captures angular velocity and acceleration at the mounting point, in which posture and transition activities can be easily distinguished.^[165,166] After processing, some important gait parameters,^[167] like foot ground clearance, and ankle velocity, can be obtained. The integration of plantar pressure sensors, EMG, and inertial sensors provides

Table 4. Gait parameters and the related calculation methods based on insole sensing systems.

Gait parameters [Unit]	Extracted methods	Related device (with auxiliary equipment)	Accuracy
Peak pressure [kPa]	Peak detection ^[171,180,222]	Plantar pressure sensor	<2.5 kPa ^{b)} [222] Not Specific ^[171,180]
Stance interval, swing interval, and stride interval [s]	Threshold division ^[87,176,180,181]	Plantar pressure sensor	<0.07 s ^{a)} [87] <1.7% ^{a)} [176] Not specific ^[180,181]
Center of pressure [m]	LASSO and SVR ^[179] Weight average ^[87,89,157,158]	Plantar pressure sensor (with IMU) Plantar pressure sensor	6.32 ± 3.09 ms ^{a)} [179] <13.8 mm ^{a)} [157] <10 mm ^{a)} [158]
Cadence [steps/min]	Threshold division ^[87,180]	Plantar pressure sensor	Not specific
Ground reaction force [kPa or N]	Summarization ^[87,89,156]	Plantar pressure sensor	<106.1 N ^{a)} [87] <68.1 N ^{b)} [89] within ±5% full scale ^[156]
	SVM ^[154]	Plantar pressure sensor	0.93 kPa ^{a)}
	Regression model ^[160]	Plantar pressure sensor	38.43 N ^{a)}
	Tenfold cross-validation process ^[161]	Plantar pressure sensor	<15.4 N ^{b)}
Stride length [m]	Threshold division ^[176–178,180,181]	IMU device (with plantar pressure sensor)	1.83 ± 0.80 cm ^{a)} [176] 1.66 ± 0.18 cm ^{a)} [177] Not specific ^[178,180,181]
	LASSO and SVR ^[179]	IMU device (with plantar pressure sensor)	1.7 ± 0.11 cm ^{a)} (LASSO) 1.59 ± 0.14 cm ^{a)} (SVR)
Foot ground clearance [m]	LASSO and SVR ^[179]	IMU device (with plantar pressure sensor)	0.42 ± 0.12 cm ^{a)} (LASSO) 0.36 ± 0.11 cm ^{a)} (SVR)
Walking speed [m/s]	Threshold division with attitude solution ^[176,177,181]	IMU device (with plantar pressure sensor)	(1.74 ± 0.83) × 10 ^{−2} m/s ^{a)} [176] (2.67 ± 0.38) × 10 ^{−2} m/s ^{a)} [177] Not specific ^[181]
	LASSO and SVR ^[179]	IMU device (with plantar pressure sensor)	(1.7 ± 0.20) × 10 ^{−2} m/s ^{a)} (LASSO) (1.55 ± 0.14) × 10 ^{−2} m/s ^{a)} (SVR)
Joint angle [°]	Attitude solution ^[181]	IMU device (with plantar pressure sensor)	<1°
Energy expenditure [METs]	Branch algorithm ^[9]	IMU device (with plantar pressure sensor)	0.69 METs ^{a)}

^{a)}With root mean square error (RMSE) ± standard deviation (SD); ^{b)}With mean absolute error (MAE).

more comprehensive gait information for medical analysis than one sensor alone. Therefore, it has a promising future for massive usage-this section reviews the plantar pressure-centered gait detection system with IMU and EMG supplements.

5.1. Plantar Pressure with EMG

As explained above, the envelope of EMG provides the information on the muscle activation electric signal, which makes

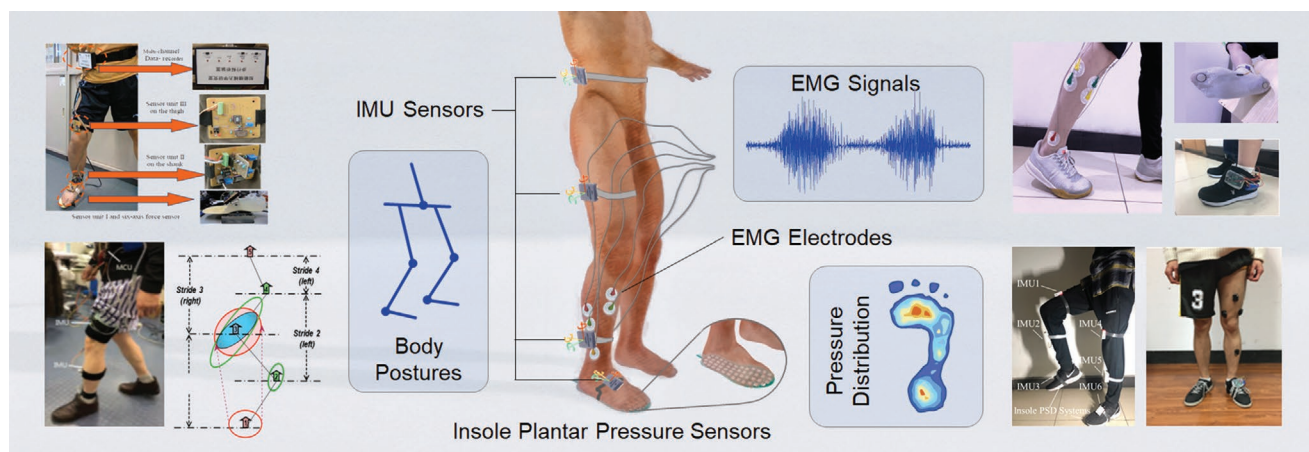


Figure 21. A scheme of insole gait monitoring system consisting of IMU, EMG, and FSR sensors. Reproduced with permission.^[166] Copyright 2020, Hindawi. Reproduced with permission.^[174] Copyright 2017, IEEE. Reproduced with permission.^[229] Copyright 2018, IEEE.

monitoring of the pathological features in muscular activation possible. However, the EMG signal lacks indicators of limb movement. Hence gait information is widely used to compensate for this. With the combination of plantar pressure, EMG signal can be analyzed at different stages of the walking cycle, which is especially significant to evaluate the relationship between muscle activation source and the final performance results (e.g., plantar pressure distribution and lower limb movement).

In 2011, D. Bojanic et al. extracted EMG characteristics of four major muscles (vastus medialis, hamstrings, tibialis anterior, and gastrocnemius medialis) from children with cerebral palsy (CP). An FSR is placed on the front of the foot to identify the swing or stance phase of the gait cycle (children with CP are toes that first touch the ground). The envelope of the EMG signal of the CP group is extracted to compare the average value with all correct classification result of the healthy group. Besides, a SIM parameter is proposed for describing the similarity of the discrete gait-related pattern of as subject under examination to the discrete norm.^[168] At the same year, A. Boschman et al. used two FSRs (placed in heel and toe) to divide the gait stage (seven-stage gait model) and measure four EMG sensors on the thigh (rectus femoris, vastus medialis, vastus lateralis, and biceps femoris) to record the RMS of EMG signal for amputee gait classification.^[169] Then in 2016, pressure-time integral, peak pressure, time to peak pressure, contact area, contact time, and CoP of patients with chronic ankle instability (CAI) are extracted by R. Koldenhoven et al. Four EMG sensors are used to collect the RMS amplitudes of muscles (tibialis anterior, peroneus longus, medial gastrocnemius, and gluteus medius). The study demonstrates a sizeable lateral deviation in the CoP of patients with CAI, and they have higher gluteus medius EMG amplitudes during the final 50% of stance and first 25% of swing.^[170] In 2020, S. Gao et al. use two FSRs and two EMG recorders on the lower limbs, extracted twelve EMG features and nine GRF features for lower limb rehabilitation, and reached the 96.8% identification accuracy of walking terrains.^[171]

5.2. Plantar Pressure with IMU

In general, the IMU module (including the accelerometer and gyroscope) can provide the biomedical information of kinematics for gait-phase division, movement analysis, and joint angle evaluation.^[152,172,173] The combination of the plantar pressure sensors allows the researchers to obtain some essential parameters like energy expenditure for sports damage recovery, and improves the estimated accuracy, and therefore be able to get more inclusive information for gait analysis.

In 2007, an IMU and FSRs were used to detect the gait phases, including initial contact, loading response, mid stance, terminal stance, pre-swing, initial swing, mid-swing, and terminal swing. The estimated results have good consistency with force platform and motion analysis system, which is consistent with the medical model.^[174] In 2008, M. Chen et al. reported a method for measuring angular velocity and acceleration using an IMU, and installed four FSRs (arranged at first metatarsal head and the position between the fourth and fifth metatarsal heads, rear and inside portion of the heel) on the

insole and a bending sensor to obtain pressure and flexion information. Four kinds of gait abnormalities (toe in, toe out, over-supination, and heel walking) can be quantified with the obtained data.^[175] In 2010, N. Sazonova et al. used the acceleration, pressure, and post/activity data from the accelerometer and pressure sensors to estimate the energy consumption during a walking event with the branch algorithm. And the RMSE of energy loss is 0.69 Mets.^[9] In 2016, by using the Sole-Sound system, J. Montes et al. analyzed the walking patterns of spinal muscular atrophy, including stride length, speed, etc., with the RMSE ($\pm$ SD) of 1.83 ($\pm$ 0.80) cm in stride length and 1.74 ($\pm$ 0.83) cm s⁻¹ in speed.^[176]

In 2017, H. Zhang et al. studied the human behavior during walking and running, and obtain the stride length (RMSE: 1.66 cm for walking, 1.91 cm for running), foot ground clearance, foot trajectory, cadence, single and double support times, symmetry ratio, CoP, and velocity (RMSE: 1.65 cm s⁻¹ for walking, 2.67 cm s⁻¹ for running) by direct calculation (weight average and threshold division) using 8-cell piezoresistive sensors and a 9 degrees of freedom (DOF) IMU. The subject-specific calibration model based on ten times cross-validation for improving the estimation accuracy is also reported.^[177] In 2018, F. Arafsha et al. report an insole platform with twelve FSRs and an IMU, which realizes wireless transmission and real-time display of data. It can be used for stride length estimation and walking direction recognition in everyday utilization, but the comparison with the reference device has not been provided.^[178] In the same year, Zhang et al. collect the raw data from a plantar pressure sensor and an embedded IMU. Two kinds of multivariate linear regression techniques, multivariate linear regression with least absolute shrinkage and selection operator (LASSO) and support vector region (SVR), are all used to calculate accurate walking parameters such as stride length (RMSE < 2.04 cm, SD < 0.15 cm), velocity (RMSE < 2.2 cm s⁻¹, SD < 0.19 cm s⁻¹), foot clearance (RMSE < 0.52 cm s⁻¹, SD < 0.14 cm), at different speeds of subjects.^[179] In 2019, S. Potluri et al. use 64 pressure sensors and an IMU to obtain the gait features like normalized peak pressure, stance ratio, step time, cadence, and the change of joint angles. The information is then used to classify the abnormal gait (Hemiplegic gait, Parkinson's gait, and sensory ataxic gait stimulated by healthy people).^[180] In the same year, Mahalingam et al. introduce a wearable-system integrated with an IMU and nine FSRs to calculate step velocity, step time, stride length, foot plate angle, and ground at the stage of heel strike and toe-off. The results show that after eight months of recovery from lower limb surgery, these five features resemble the healthy person's, which matches the surgery's rehabilitation process.^[181] The above works have been summarized in Table 4.

6. Applications of Gait System in Disease Monitoring and Analysis

Medical institutions have utilized both insole plantar pressure monitoring systems and their versions of multi-sensory integration. However, there are still two limitations preventing these devices from reaching their full potentials. First, even the detection and feature extraction can be obtained



Figure 22. The features of interest and algorithms used for disease classification.

automatically by the device itself, the analysis process still requires the healthcare professionals to carry out, and whose number is not adequate compared with the patients. Second, data detection is solely executed at the medical institutions, and the efficiency suffers from the small dataset and patients' nervousness. To address these challenges, intelligent algorithms are developed for automatic disease analysis as shown in Figure 22.

With the development of information technology, researchers demonstrate that it is possible to realize an accurate diseases or symptoms diagnosis with the help of advanced IT methods like AI and machine learning, which can help deal with the challenges of processing an overwhelmingly large number of data due to its natural generalization and capabilities of noise tolerance. In this section, both the diseases or symptoms analysis realized by AI and machine learning in both sole plantar pressure sensor and plantar pressure-centered multisensory fusion systems are presented.

6.1. Disease or Symptoms Analysis Solely based on Plantar Pressure

The first attempt to use AI to analyze plantar pressure information is given in 2010^[182] by using an insole pressure sensor array with a scanning frequency at 400 Hz. During operation, the peak pressure of each scan is recorded. After a period, the statistic results of the recorded pressure data are obtained as a form of distribution curve (percentage vs. value), which is compared with the normal distribution curve. A regression factor (RF) is proposed to quantify the similarity of the curves. RF approaching 1 indicates the patient is close to recovery. Through this method, the normal load and pathological load before and after surgery (Charcot's Neuroarthropathy) are successfully distinguished.

In 2012, S. Lee and J. Lim extracted vertical GRF of eight pressure sensors and total vertical GRF of both limbs as features, and use a neural network with weighted fuzzy membership functions (NEWFM) to classify Parkinson's gait and normal gait. The classification accuracy is 77.33%.^[183] Later in 2013, eight FSRs and sparse representation-based classification algorithms were utilized to classify the Parkinson's gait and normal gait. The features of interest are the whole vertical GRF of both limbs and vertical GRF beneath both limbs' heels

and toes (LRHT). The label consistent k-means singular value decomposition (LC-KSVD) shows the highest classification rate at 83.44% compared with 81.53% of SVD and 76.74% of sparse representation-based classification (SRC) with K-SVD trained dictionaries (SRC- $\dagger$).^[184] In 2014, Z. Li et al. compared the accuracy of twice the Bayesian standard classification method and SVM in the classification of amyotrophic lateral sclerosis (ALS), Huntington's disease (HD), and PD with the healthy control group, which is based on seven FSRs and an acceleration sensor placed insole. SVM shows a higher classification accuracy than the Bayesian standard classification. Compared with other diseases, ALS shows a greater difference than the healthy group (93.86%), and the other two conditions' accuracy is more than 85%. However, ALS, PD, and HD are not distinguished obviously from each other, possibly due to their common characteristics to some extent.^[185] In 2015, R. Alkhatib et al. classified the Parkinson's gait from normal gait with K-nearest neighbors (kNN) algorithm based on eight pressure values of both limbs. This study achieves the classification accuracy of 83%.^[186]

In 2017, E. Park et al. placed three FSR pressure sensors on the big toe, mid-late plantar, and heel-strike. By utilizing the calculated changes in walking speed and walking frequency before and after drinking, the random forest method achieves 86.2% accuracy of drunk gait classification.^[187] In the same year, a stacked 2-dimensional and 1-dimensional convolutional neural network is used to classify the publicly available plantar pressure dataset of PD and healthy subjects of similar ages by N. Hoang et al. This model achieves the highest accuracy of 88.7% compared with label consistent k-means singular value decomposition algorithm (LC-KSVD, 83.4%), kNN (83.0%) and wavelet + neural network with fuzzy membership functions (77.3%).^[188] Besides, J. Ramirez et al. presented fuzzy cognitive maps trained by a genetic algorithm (GA-FCM) against a multilayer perceptron neural network (MLPNN) based on the pressure of 14 plantar pressure sensing regions. The classification accuracy of flat and cavus feet is 91% and 87%, respectively.^[189]

6.2. Disease or Symptoms Analysis Based on Insole Device-Centered Multisensory Fusion

In Section 5, we have discussed using plantar pressure-centered multisensory fusion systems to extract various gait features. Based on that, some researchers use the extracted features as

Table 5. Related works at diseases or symptoms analysis based on gait information.

Diseases/symptoms	Algorithms/methods	Parameters of interest	Accuracy score
Classify Parkinson's gait and healthy gait ^[180,183–186,188]	Stacked LSTM ^[180]	Normalized peak pressure, stance ratio, step time, cadence, and joint angles velocity	98%
	NEWFM ^[183]	Value of eight pressure sensors, and vertical GRF of both limbs	77.33%
	LC-KSVD, SVD, and SRC [†] ^[184]	Vertical GRF of both limbs, pressure beneath heels and toes of both limbs	83.44%
			81.53%
			76.64%
	Bayesian Standard Classification and SVM ^[185]	Stride interval, swing interval, stance interval of both feet, and double support interval	84.36%
Classify amyotrophic lateral sclerosis, HD, and healthy gait ^[185]	kNN ^[186]	Value of eight pressure sensors	83%
	A stacked 2D and 1D CNN ^[188]	Vertical GRF of both limbs	88.7%
			83.34%
Classify post-surgery of Charcot's neuroarthropathy, after reconstruction and healthy people ^[182]	Bayesian Standard Classification and SVM	Stride interval, swing interval, stance interval of both feet, and double support interval	91.57%
Classify Hemiplegic gait, sensory ataxic gait, and healthy gait ^[180]	Time scale normalization and multiple regression	Peak pressure of each pressure frame during a walking cycle	RF ^{a)} used for classification
Classify flat foot and cavus foot ^[189]	Stacked LSTM	Normalized peak pressure, stance ratio, step time, cadence, and joint angles velocity	98%
Classify CP gait and healthy gait ^[168]	GA-FCM and MLPNN	Normalized plantar pressure in 14 regions (toe 1st to 5th, metatarsal joint 1st to 5th, lateral midfoot, medial midfoot, lateral heel, and medial heel.)	91.1%
			87.2%
Classify four abnormal gait including toe in, toe out, over-supination, and heel walking and healthy gait ^[175]	Statistical analysis	The average of EMG envelope	SIM ^{a)} used for classification
	PCA with SVM	Angular velocity, acceleration, pressure, and flexion information	90.75%

^{a)}The extracted parameter with a high correlation of interested disease.

the model input for symptoms classification, and promising results are obtained. In this part, the classification algorithms and its accuracy in plantar pressure-centered multisensory fusion systems are presented.

In 2008, FSRs and IMU are used to extract gait features using PCA, then the SVM classification model is trained to classify four abnormal gaits (toe in, toe out, over-supination, and heel walking) and reaches the average classification accuracy of 90.75%. Besides, the accuracy of merely using insole sensors is 7.1% higher than using IMU only, and around 3.5% lower than using both sensing dimensions. The result demonstrates that integrating multiple sensors can obtain more comprehensive information on gait, which is valuable for diseases and symptoms analysis.^[175] In 2011, through using an FSR sensor and four EMG sensors, SIM was extracted to analyze the subjects' walking patterns. As a result, SIM decreases mainly in all measured muscles of the CP group, which means the loss of athletic ability.^[168] In 2019, S. Potluri et al. proposed a classification method using a stacked long short-term memory (LSTM) model based on 64 pressure sensors and an IMU. Four gait abnormalities (Hemiplegic gait, Parkinson's gait, and sensory ataxic gait stimulated by healthy people) are classified with the prediction errors of 2%.^[180] The above works are all summarized in Table 5.

Although reported work shows the high accuracy of disease identification using some AI algorithms, the input features

should be considered for their redundancy. Among the extracted features, some have a more reliable connection to specific diseases compared to their counterparts. Hence, if all elements are used to classify activities, then the model redundancy will be high,^[190] lowering the classification accuracy. Therefore, the selection of gait features should be carefully made before model training for getting a more accurate and useful model.

7. Challenges and Future Development

Though remarkable research outcomes have provided various techniques for insole sensor design, gait parameters reconstruction and extraction from insole gait monitoring systems based on pressure sensors as well as their multi-sensory integration versions, and automatic disease classification algorithms, insole gait monitoring systems have not been successfully applied in daily lives yet. This phenomenon could be explained in various aspects. Nevertheless, three key factors, which are the raw data characteristics, energy consumption, and the requirement of large sensor datasets for disease emulation, may play a significant role. In this section, these three factors and their influences on the development of plantar pressure-based insole gait monitoring systems are discussed, and we review possible solutions of these challenges based on current knowledge and state of the technology.

7.1. Challenges

7.1.1. Raw Data Characteristics

The quality of the collected raw data from sensors determines the final accuracy of further processing and disease identification. Unfortunately, raw data's reliability is mainly affected by two efforts: the slip of sensors during walking and multiple disease fusion.

In terms of the former, after raw data collection, all the following signal processing is based on the assumption that the relative position between sensors and the human body remains the same during walking. However, in real-life scenarios, the shoes and feet are not glued. Since gait sensors are placed at a fixed area attached to shoes, the slip between insole and foot is inevitable, resulting in a reduction of gait registration accuracy. Therefore, determining foot position while detecting is highly expected for a more accurate data collection.

As to the latter, the gait information is contributed by multiple diseases, whose weighing factors are unknown. Therefore, it is quite hard to single out one disease's contribution, which is vital for sole disease diagnosing and monitoring. Hence, for better performance improvement of the classification model, it is essential to work out a way to determine sole disease's influence on the gait information.

7.1.2. Energy Consumption

The wearable gait detection system often needs to work independently, its working time is mainly determined by battery capacity and power consumption rate. Thus, for longer battery life and better user experience, one can either increase battery capacity, reduce power consumption, or charge the battery during working.^[191] Commercial batteries are not suitable for wearable applications that require frequent bend, large extrusion, and long working time because of their hard texture and low energy density ($<265 \text{ Wh kg}^{-1}$).

One way to reduce the module power consumption of electronic circuits is by using low-power discrete components instead of conventional modules. However, the energy loss in wire resistance and distribution parameters is inevitable.

Besides, solar energy and bioenergy generated technologies can significantly prolong battery life through charging the battery during work. However, solar technology requires a stable illumination environment, and the optical-electrical efficiency decreases significantly in indoor or dark environments. The electric energy generated from bioenergy is less than 10 mW, which can only support a single circuit module or a circuit system with an application-specific, low-power design. The materials with high energy conversion efficiency show great potential in increasing the output power. However, the compatibility with wearable systems and the fabrication process needs to be further studied and optimized.

7.1.3. Collection of Large Sensor Datasets for Disease Emulation

Generally speaking, the model training and evaluation algorithms require a large dataset provided by multiple sensors

from various periods and individuals. However, the mass production of instrumented shoes has not yet been realized, and experimental objects are still a small part of the targeted crowd, making the obtained dataset unable to represent the whole, possibly undermining the algorithm's accuracy when applied to the entire desired population.

Besides, building a large dataset, many other parameters like location, weather, etc. around the subjects can be added as additional information, helping medical institutions better understand the disease progression, cause, and give a more personalized rehabilitation program to patients.

7.2. Future Development

To address the current challenges in insole pressure applications, possible solutions have been developed based on the current knowledge and state of the technology to improve raw data quality, prolong the working time and build large datasets as follows:

7.2.1. High Data Quality and Intended Information Extraction

The raw data quality suffers from the device slip phenomenon and multiple disease fusion. The AI method is a feasible scheme to detect the slip phenomenon by identifying the abnormal pressure pattern. This method has not been utilized in the insole field yet but has been successfully applied to tactile sensing devices, inspiring the insole pressure sensing field. For example, Y. Akiyama et al. used two optical sliding sensors and four force sensors with a 3D motion capture system to model the slip phenomenon of a wearable robot in the sitting-standing process. The author pointed out that the interaction force and torque will increase when a slip occurs, and the slip phenomenon can be determined by monitoring the interaction force mode near the contact point.^[192] Based on this, A. Begalinova et al. fabricated a smart glove with fingertip pressure sensor and 10 degrees of freedom inertial sensor. The sliding and non-sliding events classification was completed with 89% accuracy utilizing the pressure data of the time series and long short-term memory networks.^[193]

To extract the effect of a specific disease on patients' gait from multiple diseases fusion, it is necessary to track the gait of different patients for a long time and continuously analyze the gait of the same patient. Then, the most typical characteristics associated with a specific disease can be evaluated based on the doctors' diagnosis and the gait changes under the effect of a single disease or multiple diseases. This intended information extraction method has preliminarily demonstrated its feasibility in the analysis and treatment of cardiovascular diseases. Through numerical simulation, O. Mazumder et al. utilized the hemodynamic model to establish a biventricular digital twin model and successfully extracted the characteristic photoplethysmogram signal of coronary artery disease (a symptom of atherosclerosis). Besides, patients with coronary artery disease were successfully distinguished from other cardiovascular patients based on the features extracted from the photoplethysmogram signals.^[194]

7.2.2. Prolong the Working Time

To increase the battery lifetime of the insole system, having a large battery capacity, low power consumption, and charging the battery during working are three optimal solutions.

Large Battery Capacity: Using electrode materials with higher energy density (e.g., Li-S battery and zinc-air battery) to improve the battery capacity per unit volume has become a research hotspot in recent years. To this extent, many flexible batteries with high energy densities have been developed. Jun-Woo et al. reported a flexible high energy density lithium-sulfur (Li-S) battery based on fibrous sulfur catalysts and separators. It has excellent electrochemical performance stability and energy density up to 443 Wh kg⁻¹.^[195] Liu et al. fabricated a rechargeable liquid zinc-air battery based on a flexible nanoporous carbon fiber film through pyrolyzed of electrospun polyimide. A high energy density of up to 776 Wh kg⁻¹ was achieved.^[196] These two new batteries can improve endurance time, provide better mechanical properties, and are more suitable for the insole application. However, a massive production method for battery materials with high energy density and good mechanical properties is not yet offered.

Low Power Consumption: In battery-powered wearable systems, leakage current has become the main component of total power consumption.^[197] By improving the circuit structure (e.g., stack transistors^[198]) and introducing switch control logic to reduce the leakage current of non-work components (e.g., metal oxide semiconductor), the power consumption of integrated chips (operational amplifiers, analog-to-digital converter, and microcontroller) can be effectively reduced. J. Zhang et al. utilized the two-stage current reuse technique to embed the driving unit of the class-AB output stage with the amplifier input stage to improve the supply current utilization efficiency. The optimized output power of the fabricated amplifier is 320 nW.^[199] W. Mao et al. proposed two stacked transistor structures with internal capacitance to reduce the leakage current of the analog-to-digital converter. Combined with multi-segmented analog-to-digital converter architecture and hybrid switch scheme, the operating power consumption is only 110 nW at 1 kS s⁻¹ sampling rate, 12-bit resolution, and 1.0 V voltage.^[200] N. Ickes et al. optimized the memory access process with two on-chip low-voltage static random access memory (used for instruction and data storage, respectively). The microinstruction cache is optimized to improve the instruction hit rate to achieve a low energy consumption. The prepared 32-bit microcontroller provides I²C, SPI, and parallel communication interfaces with the power consumption of 3.44 nW at the main frequency of 82.5 MHz and the working voltage of 1.2 V.^[201] However, the adverse impact of wire resistance and distribution parameters on power consumption cannot be compensated by the discrete low-power components, which still consume additional energy.

Additionally, integrating discrete functional modules into a single application-specific integrated circuit and optimizing for the specified application can achieve a low power consumption at the system level. For example, P. Ahmmedet et al. reported an application-specific integrated circuit of ultra-low-power photoplethysmogram detector based on CS, which integrates the front-end analog circuit and a digital back-end processor. The average power consumption of only 172 μW.^[202]

Charge the Battery during Working: Supplementing battery energy during physiological parameter detection can be realized by using solar power, mechanical power conversion (e.g., the direct piezoelectric effect methods^[203]), and biothermal power.^[204] Among them, the solar energy harvester is the conventional charge-in-work method utilized in the wearable device field. However, the solar power-based conversion method suffers from low photoelectric conversion efficiency (around 14–19% for Si solar cells) due to light reflection, dynamic impedance changes with respect to light intensity, and material surface defects. Coating anti-reflective coatings on the surface (typical materials include TiAl₂,^[205] SiO₂,^[206] ZnS,^[207] Al₂O₃, Ta₂O₅-Al₂O₃^[208] and thermoplastic PU-based polymer material^[209]) can effectively solve the problem of light reflection and enhance the absorption of sunlight, so as to improve the efficiency of solar cells (the improvement is around 10–14%^[208]). For the light-intensity-dependent battery impedance, maximum power point tracking technology (e.g., incremental conductivity of temperature and irradiance compensation,^[210] duty cycle adjustable DC-DC converter based on meteorological information^[211,212]) can be implemented to match the load impedance with the solar cell. In this case, the output efficiency of the solar cell is always maintained at the maximum value (these technologies can increase the efficiency of the solar cells by 30–60%). To address the surface defects problem of conventional silicon heterojunction solar cells, the introduction of a thin intrinsic layer into the two doped layers of the heterojunction (the structure is called a heterojunction with an intrinsic thin layer, and the typical structures are graphene/MoS₂ heterostructure,^[213] crystal silicon heterojunction^[214–216]) can greatly reduce the surface defects. With a thin intrinsic layer, Panasonic achieved the world's highest photoelectric conversion efficiency (25.6%) of solar cells.^[214] Based on these three improvement measures, the efficiency of a solar panel is expected to be much higher. However, solar cell performance is greatly reduced in dark scenes (e.g., indoor environment or cloudy days).

The use of mechanical power and biothermal power can directly convert human biological signals into electrical energy, which is not limited by the environment and has a broader application prospect. As for the mechanical power-based energy harvester, in ref. [203], J. Zhao uses piezoelectric material, PVDF, to build a mechanical energy harvester for force-induced energy collection during walking. As a result, the mean output power is 1 mW. To improve the human biological-kinetic energy conversion ability, piezoelectric materials with higher energy conversion coefficient can be used (e.g., lead magnesium niobate lead titanate (PMN-PT) single crystals^[217]). However, the direct piezoelectric effort is only applicable to harvest energy at dynamic gait.^[218] As for the mechanical power-based energy harvester, thermopiles were attached to the skin or clothes to generate power. With the ambient temperature increase from 15 °C to 27 °C, the generated power is decreased from 5 mW to 0.5 mW.^[219] The generated power is enough for wireless electrocardiography^[220] but not for gait analysis like plantar pressure measurements. Although there are many available wearable energy collection schemes, how to further improve energy collection efficiency, integrate it into wearable devices, and design a low-cost and large-scale fabrication process continues to be an issue waiting to be solved.

7.2.3. Build Large Datasets

The cooperation of multinational medical systems is highly expected to establish a sizeable disease-associated dataset. International health organizations (such as WHO) take the lead in formulating laws and regulations to clarify norms and standards for protecting patient privacy, data security, and achievement distribution. At the same time, local medical institutions can popularize the research's importance and benefits to the public by leaflets and popular science videos to collect data of interest conveniently. Based on this, the medical staff can upload the plantar pressure data and the patient's medical history to establish a data-sharing platform. After huge data volume is obtained, researchers can use analysis methods (e.g., statistical analysis and AI) to evaluate the performance of diseases on plantar pressure and build the disease evaluation model. For example, the researches in refs. [180,185] used the plantar pressure dataset of patients (including neurodegenerative diseases like ALS, PD, and HD) published on PhysioNet,^[221] successfully extracted the typical abnormal parameters of patients, and trained the classification models for diagnosis. These attempts show the potential in building the relationship between abnormal gait patterns and specific diseases. In the future, the established large datasets will significantly benefit diagnosis, rehabilitation, and even early warning for diseases.

8. Conclusion

This article investigates and reviews the status of insole plantar pressure sensing techniques and their use in disease analysis. Seven mainstream sensing mechanisms, including resistive, capacitive, piezoelectric, inductive, optical, and air pressure based architectures, are explained and emphasized by criticizing their pros and cons. Sensor design considerations in terms of materials, geometry and deployment are also studied. Besides, electronics and diverse processing algorithms for manipulating sensors' outputs to transform raw pressure data to understandable gait features are introduced. Furthermore, a glance at plantar pressure centered multisensory fusion techniques is offered, together with disease analysis cared application demonstrations. Last, challenges limiting the broad and successful use of insole techniques in disease analysis and corresponding potential solutions and development trends are discussed.

The current state-of-the-art insole gait monitoring techniques illustrate a promising utilization future for disease analysis and post-operation rehabilitation. With the fast development of flexible electronics, artificial intelligent and advanced communication technologies, smart gait monitoring systems have a strong potential to become our daily used wearable systems, like smart watch, to provide us essential motion information for health and related applications.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

plantar pressure detection, gait monitoring, insole system, disease analysis

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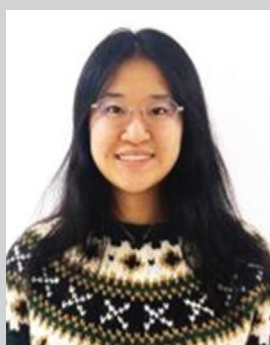
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